Screening in digital monopolies

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Economic Theories of the Information Society

Free damaging and replication

Several goods exhibit:

1. Free replication;
2. Free damaging.

↑└ taste heterogeneity (Deneckere and McAfee, 1996)

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Examples of **digital goods**:

1. Software goods;
2. Digital content;
3. Data.

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Up to 32,767 variables	-	✔	✔	✔	✔
Up to 120,000 variables	-	-	✔	✔	✔
Maximum number of observations					
Up to 2.14 billion	✔	✔	✔	✔	✔
Up to 20 billion	-	-	✔	✔	✔

Preview

1. This paper studies screening when production features free replication and free damaging.
2. The **monopoly** provision features two interdependent inefficiencies:
   - 2.1 **Productive inefficiency**: the best quality is lower than its efficient level;
   - 2.2 **Distributional inefficiency**: some buyers purchase a damaged good.
3. Extensions:
   - A no-damaging constraint makes productive inefficiency stronger;
   - **Competition** is harmful for productive efficiency and beneficial for damaging inefficiency.

Model

Model

A unit mass of buyers, each drawing a **type** $\theta \in [0,1] = \Theta$, interacts with a seller.

Type θ is privately informed about $\theta \sim F$, for twice diff. F;
$\hookrightarrow$ F is regular except in dedicated slide, $\mathbb{E}\{\cdot\}$ refers to the r.v. θ.

Type θ has payoff from **quality** $q \in \mathbb{R}_+$ and transfer $t \in \mathbb{R}$:

$$\underbrace{g(q) + \theta q}_{\text{utility } u(q,\theta)} - t,$$

for a strictly concave, increasing, and twice diff. g with $g(0) = 0$.

An **allocation** is a measurable $\boldsymbol{q}\colon \Theta \to \mathbb{R}_+$.

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An **allocation** is a measurable $\boldsymbol{q}\colon \Theta \to \mathbb{R}_+$.

The cost of allocation $\boldsymbol{q}$ is

$$C(\boldsymbol{q}) = c(\sup \boldsymbol{q}(\Theta)),$$

for a **production cost** c, increasing, strictly convex, twice diff., with $c'(0) = 0$ and $\lim_{q\to\infty} c'(q) = \infty$.

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An **allocation** is a measurable $\boldsymbol{q} \colon \Theta \to \mathbb{R}_+$.

With *separable* costs, the cost of $\boldsymbol{q}$ is

$$C(\boldsymbol{q}) = \mathbb{E}\{k(\boldsymbol{q}(\theta))\},$$

for some k (Mussa-Rosen '78.)

Efficiency

The *surplus* induced by allocation $\boldsymbol{q}$ is

$$\mathbb{E}\{u(\boldsymbol{q}(\theta), \theta)\} - c(\sup \boldsymbol{q}(\Theta)).$$

The *efficient* allocation $\boldsymbol{q}^\star$ maximizes surplus.

Proposition 1

The efficient allocation is given by $\boldsymbol{q}^\star(\theta) = q^\star$ for all θ, in which $q^\star$ is the unique quality q such that

$$g'(q) + \mathbb{E}\{\theta\} = c'(q).$$

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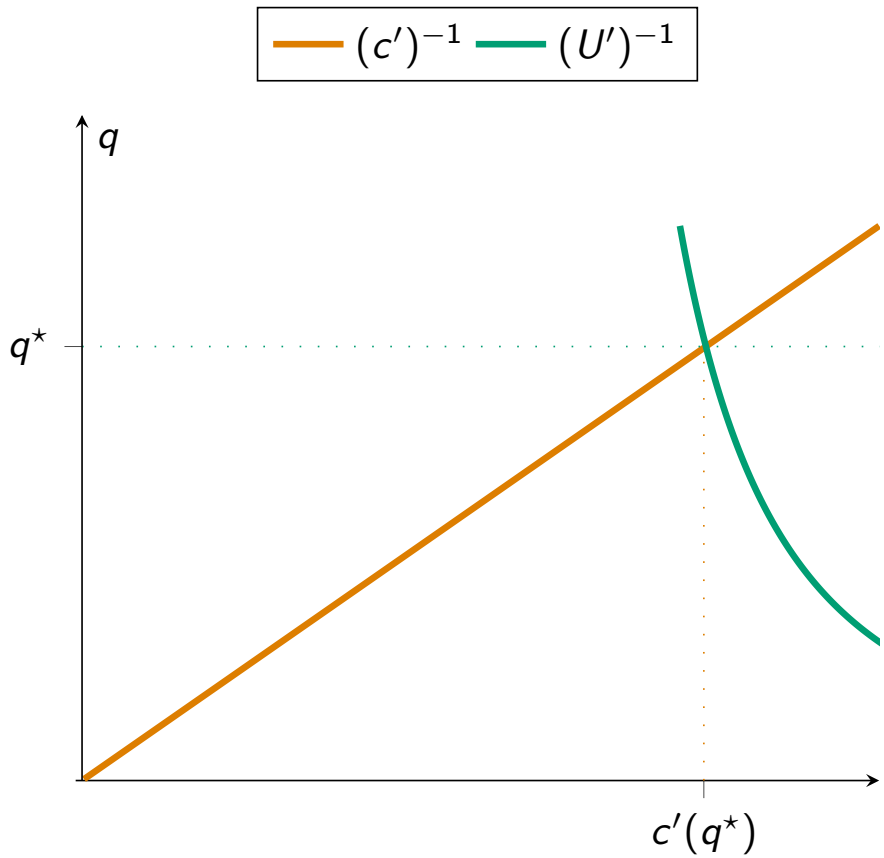
Proposition 1

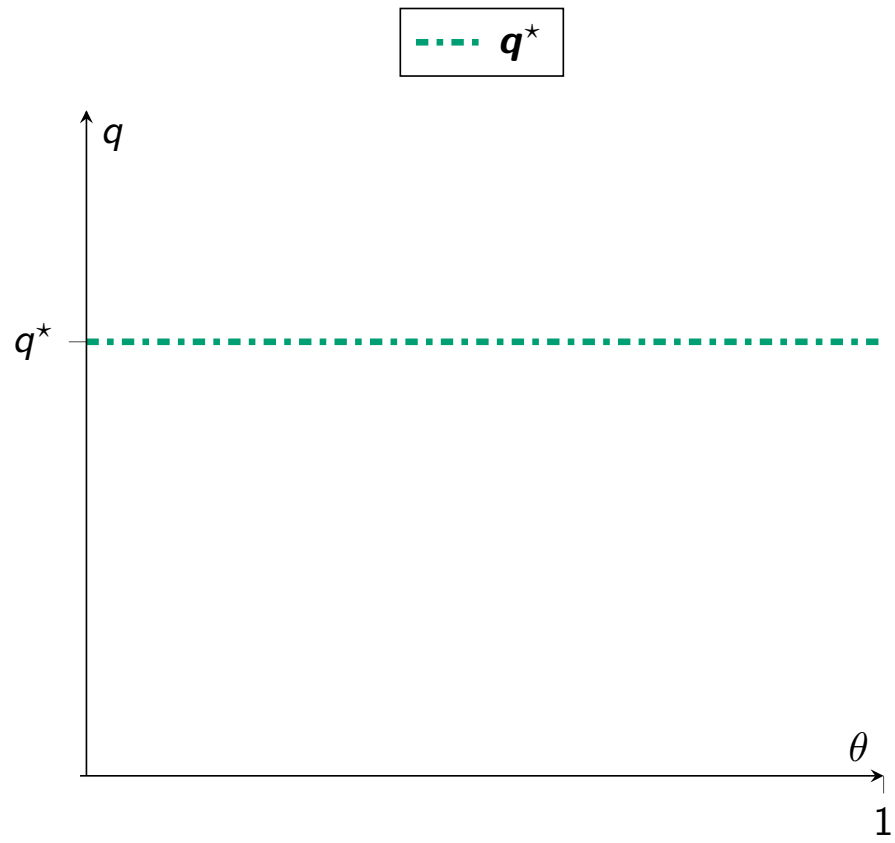
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1. Damaging is inefficient: $\mathbb{E}\{u(\sup \boldsymbol{q}(\Theta), \theta)\} \geq \mathbb{E}\{u(\boldsymbol{q}(\theta), \theta)\}$;
2. Average marginal utility equals marginal production costs.

Efficiency





The utility induced by allocation $\theta \mapsto q$ is $U(q) = g(q) + \mathbb{E}\{\theta\}q$.

Monopoly

Monopoly

The monopolist problem is:

$$(\mathcal{P}^M) \qquad \max_{\boldsymbol{q},\, t(\cdot)} \int_\Theta t(\theta)\, \mathrm{d}F(\theta) - c(\sup \boldsymbol{q}(\Theta)) \text{ subject to:}$$

$$u(\boldsymbol{q}(\theta), \theta) - t(\theta) \geq u(\boldsymbol{q}(\hat{\theta}), \theta) - t(\hat{\theta}), \text{ for all } (\theta, \hat{\theta}),$$
$$u(\boldsymbol{q}(\theta), \theta) - t(\theta) \geq 0, \text{ for all } \theta.$$

- The *monopolist* allocation $\boldsymbol{q}^M$ solves $\mathcal{P}^M$ for some $t(\cdot)$.
- Without separable costs: the monopolist problem cannot be solved via "pointwise maximization".

Monopoly

The *q-constrained problem* is:

$$(\mathcal{P}(q)) \qquad V(q) := \max_{\boldsymbol{q},\, t(\cdot)} \int_{\Theta} t(\theta)\, \mathrm{d}F(\theta) - \cancel{c(\sup \boldsymbol{q}(\Theta))} \text{ subject to:}$$

$$\boldsymbol{q}(\theta) \leq q, \text{ for all } \theta,$$

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Lemma 1 (Invest then distribute)

The allocation $\boldsymbol{q}$ solves $\mathcal{P}^M$ if and only if:

1. $\boldsymbol{q}$ solves $\mathcal{P}(q^M)$,
2. q^M solves $\max_q V(q) - c(q)$.

Monopoly

The *q-constrained problem* is:

$$(\mathcal{P}(q)) \qquad V(q) := \max_{\boldsymbol{q}} \int_\Theta \underbrace{g(\boldsymbol{q}(\theta)) + \varphi(\theta)\boldsymbol{q}(\theta)}_{\text{Virtual surplus}} \mathrm{d}F(\theta) \text{ subject to:}$$

$$\boldsymbol{q}(\theta) \leq q, \text{ for all } \theta,$$

$$\boldsymbol{q} \text{ is nondecreasing;}$$

in which $\varphi(\theta) := \theta - \frac{1-F(\theta)}{F'(\theta)}$.

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Virtual surplus maximization

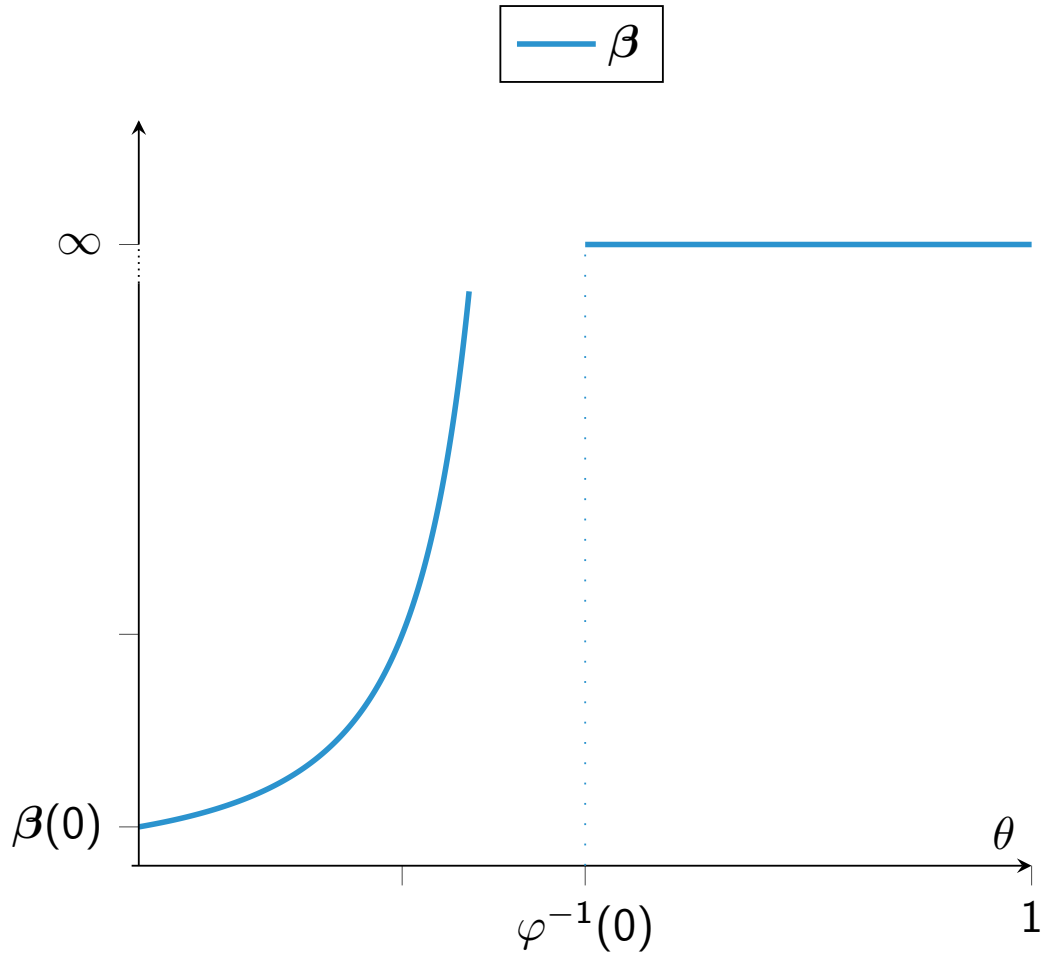
The *virtual-surplus maximizer*

$$\beta(\theta) \in \operatorname*{Argmax}_q g(q) + \varphi(\theta)q$$

satisfies:

1. $\beta(\theta) = \infty$ if $\theta \geq \varphi^{-1}(0)$;
2. β is increasing;
3. $\beta(0) > 0$ ("Inada" g).

b is the inverse of β.



Virtual surplus maximization

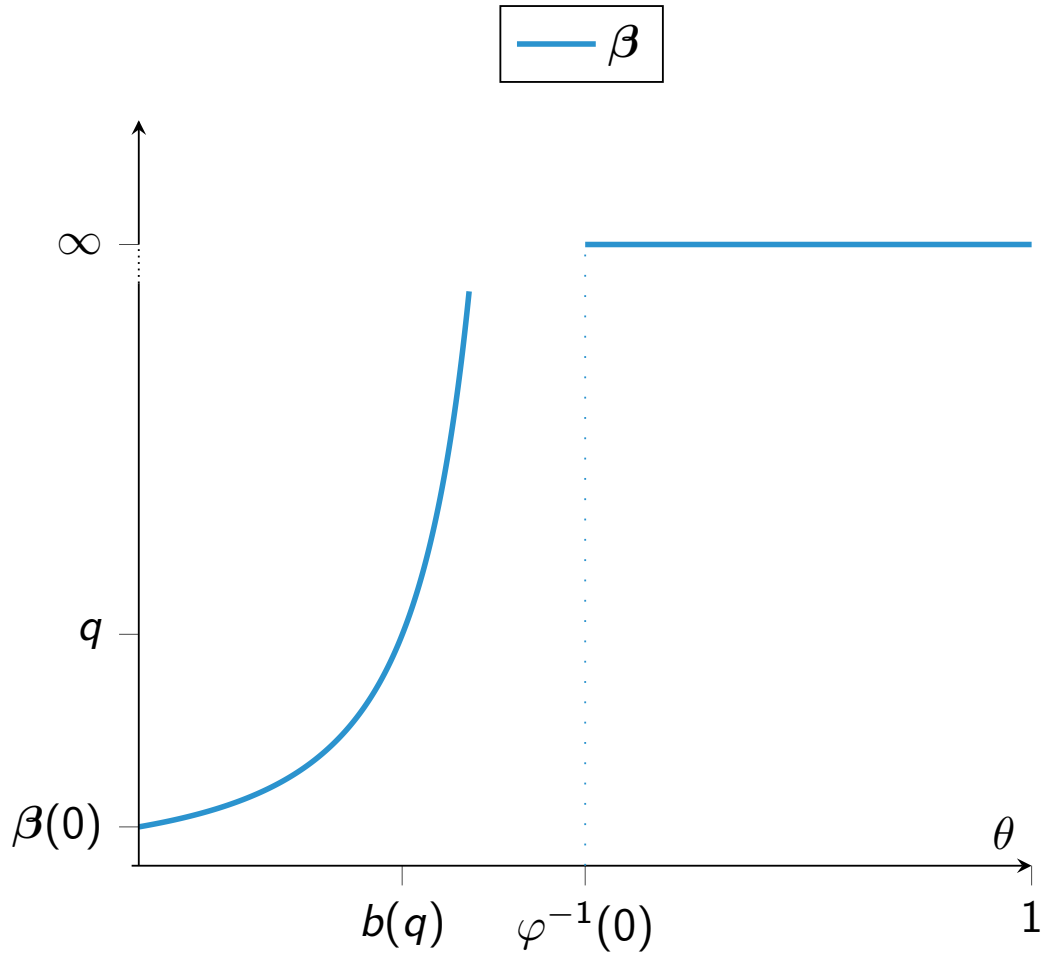
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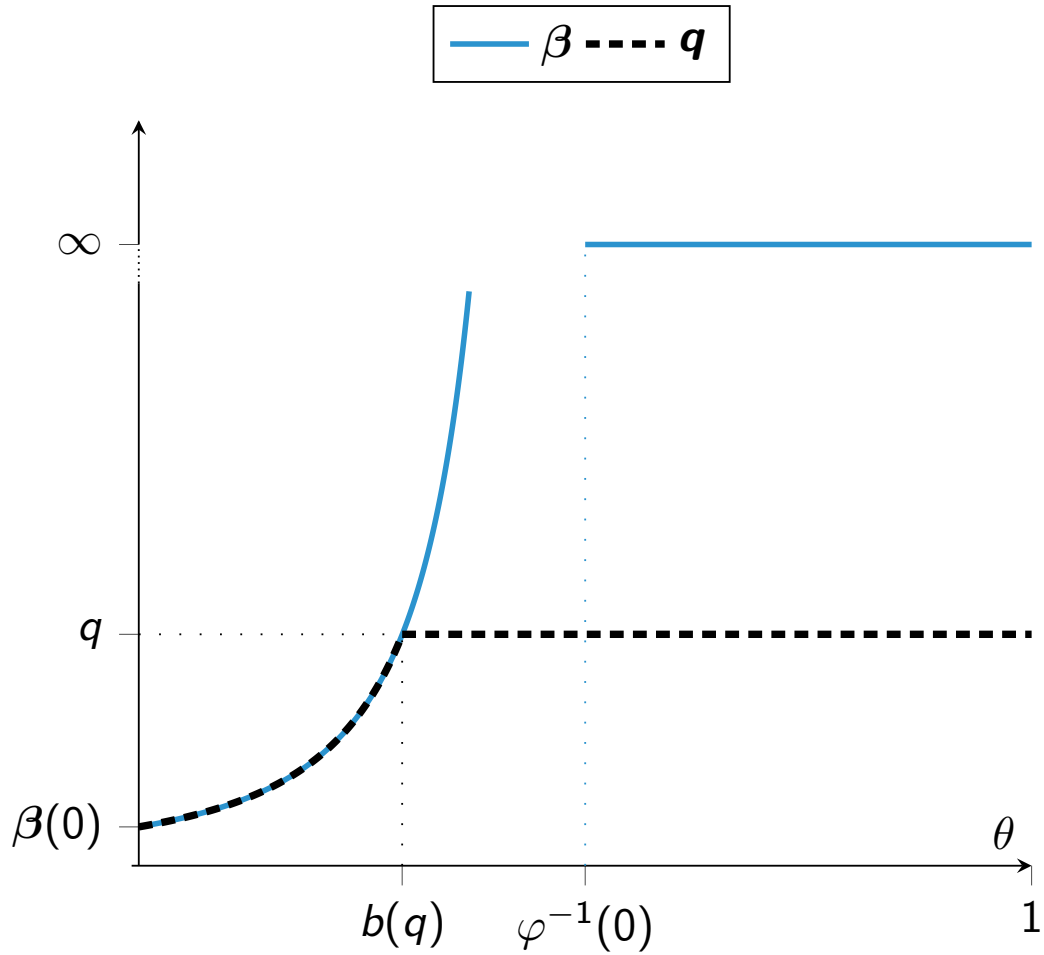
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Virtual surplus maximization

Lemma 2
Allocation $\boldsymbol{q}$ solves $\mathcal{P}(q)$ iff:

$$\boldsymbol{q}(\theta) = \min\{\boldsymbol{\beta}(\theta), q\}, \text{ for all } \theta.$$



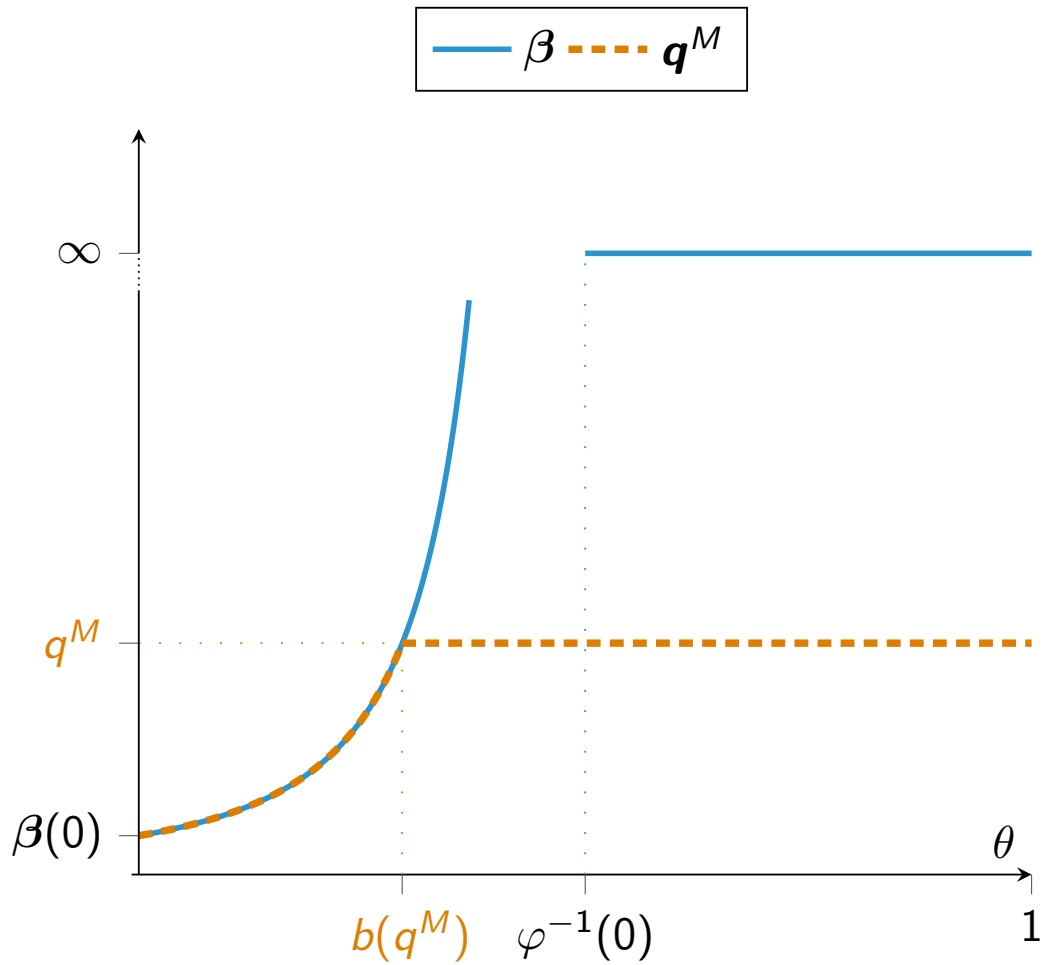
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Distributive properties of $\boldsymbol{q}^M$:

1. Bunching at the top;
2. Distributional inefficiency at the bottom or full bunching;
3. No exclusion (if $q^M > 0$.)



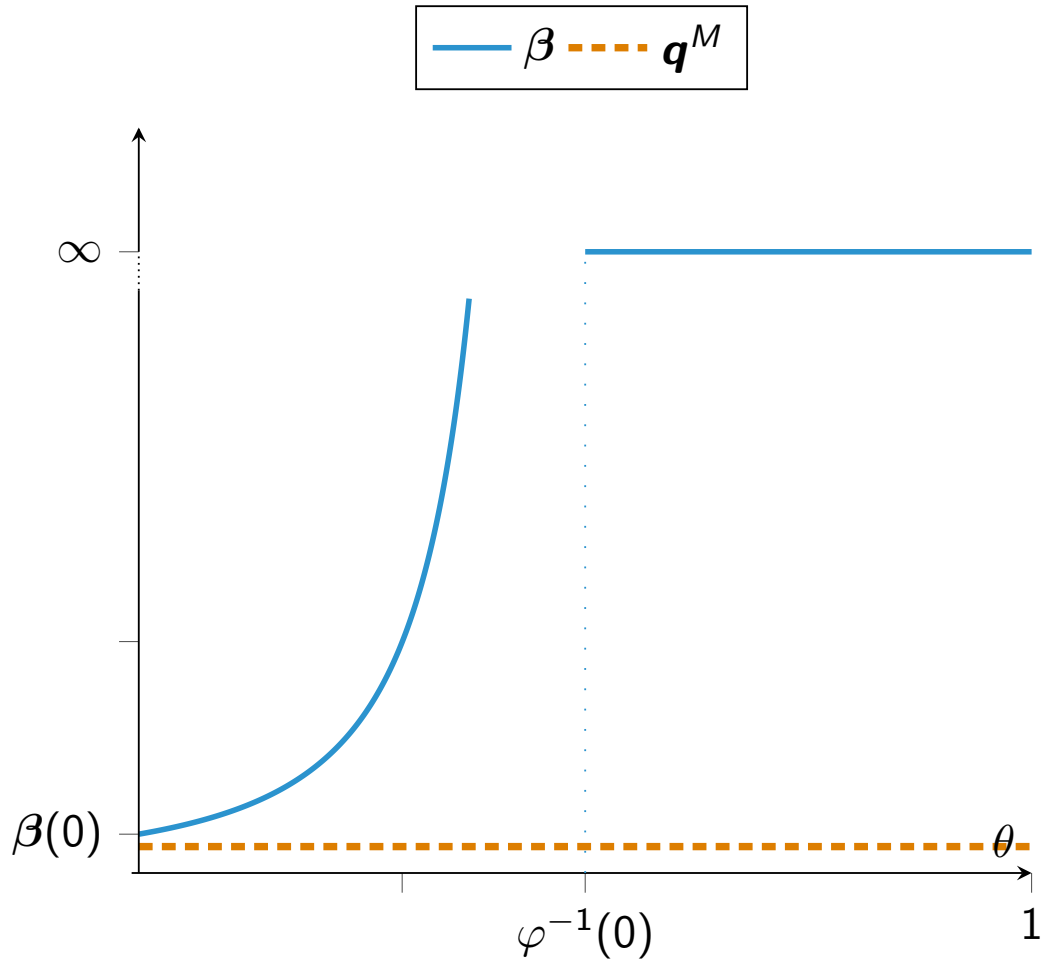
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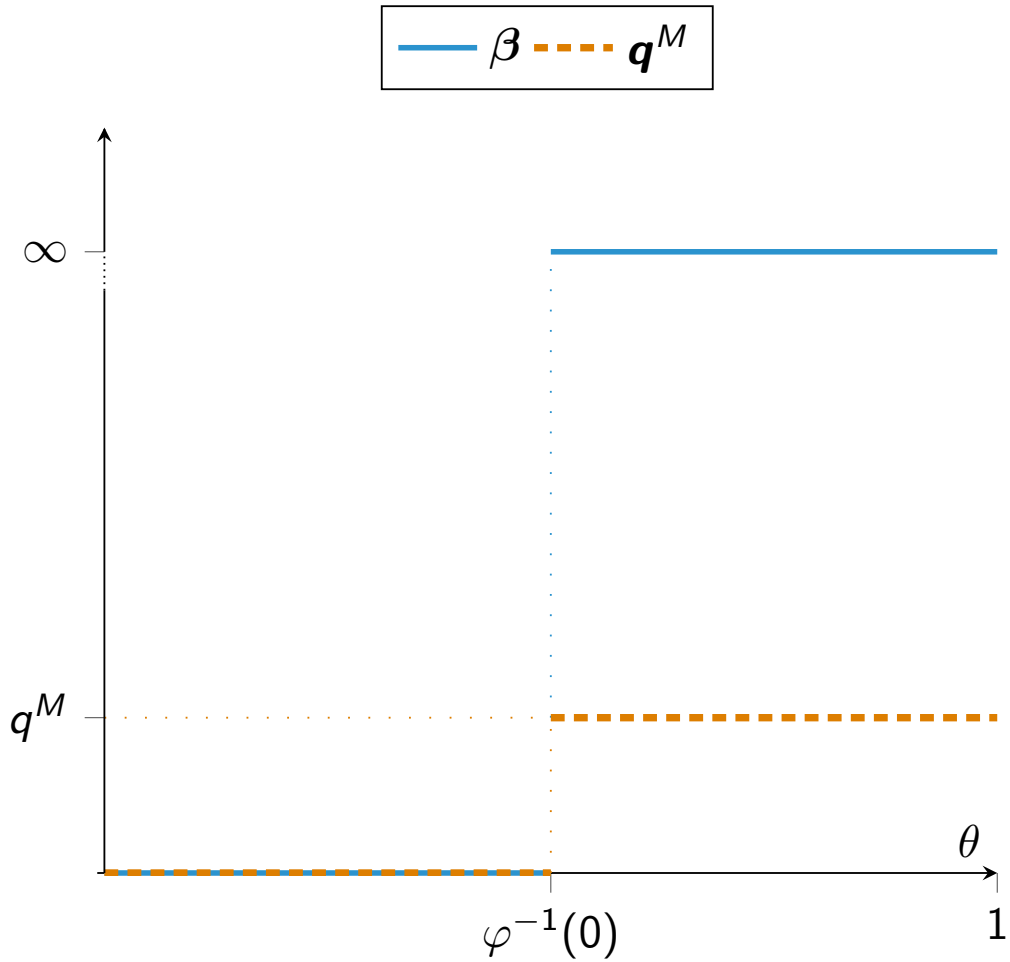
Linear preferences

Distributive properties if $g(q) = 0$:

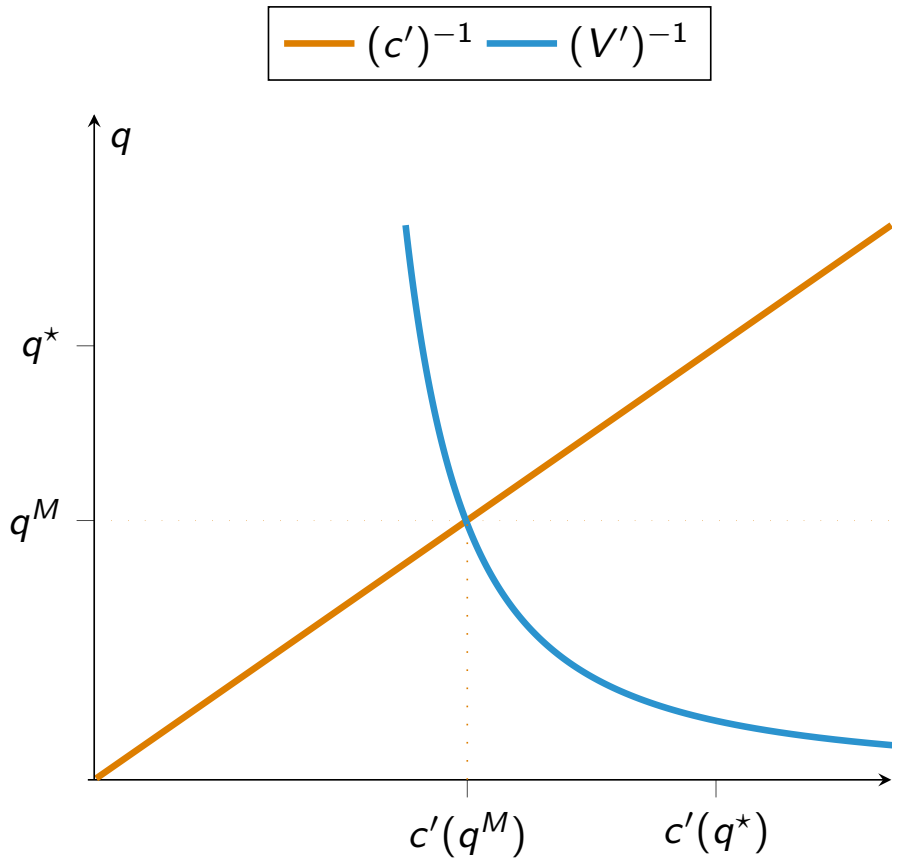
1. Bunching at the top;
   $\beta(\theta) = \infty$ for $\theta \geq \varphi^{-1}(0)$
2. Exclusion at the bottom;
   $\beta(\theta) = 0$ for $\theta < \varphi^{-1}(0)$

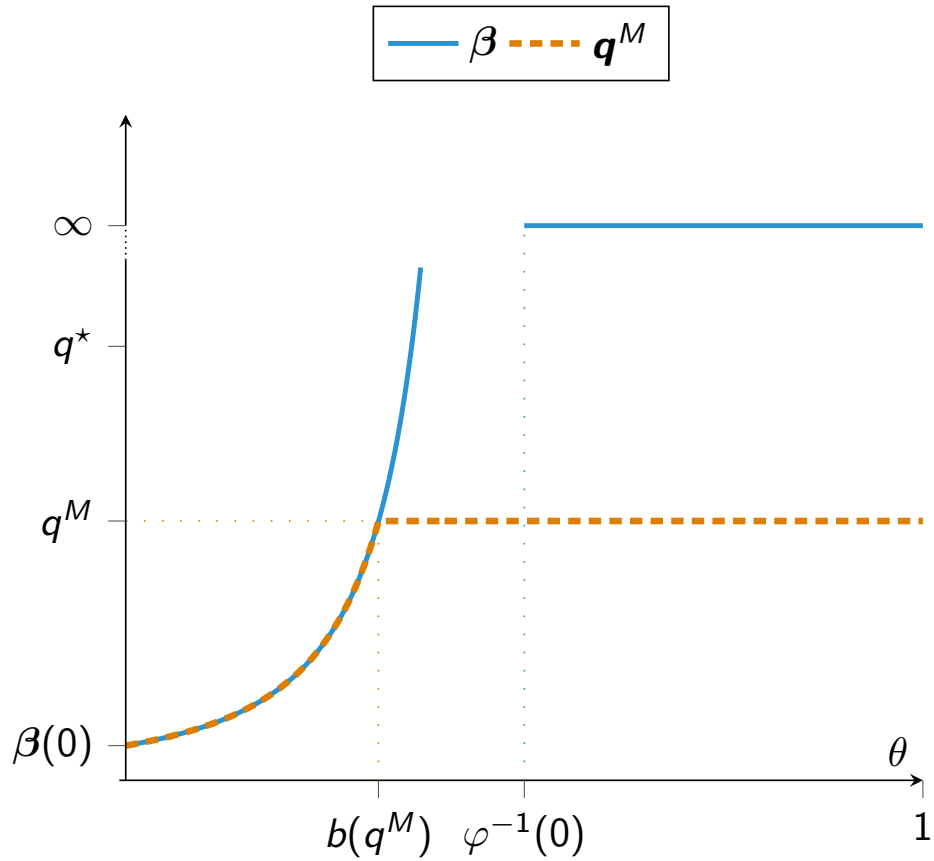
$\Longrightarrow$ **single version**.

Richness in digital markets is due solely to preferences.



The monopolist allocation

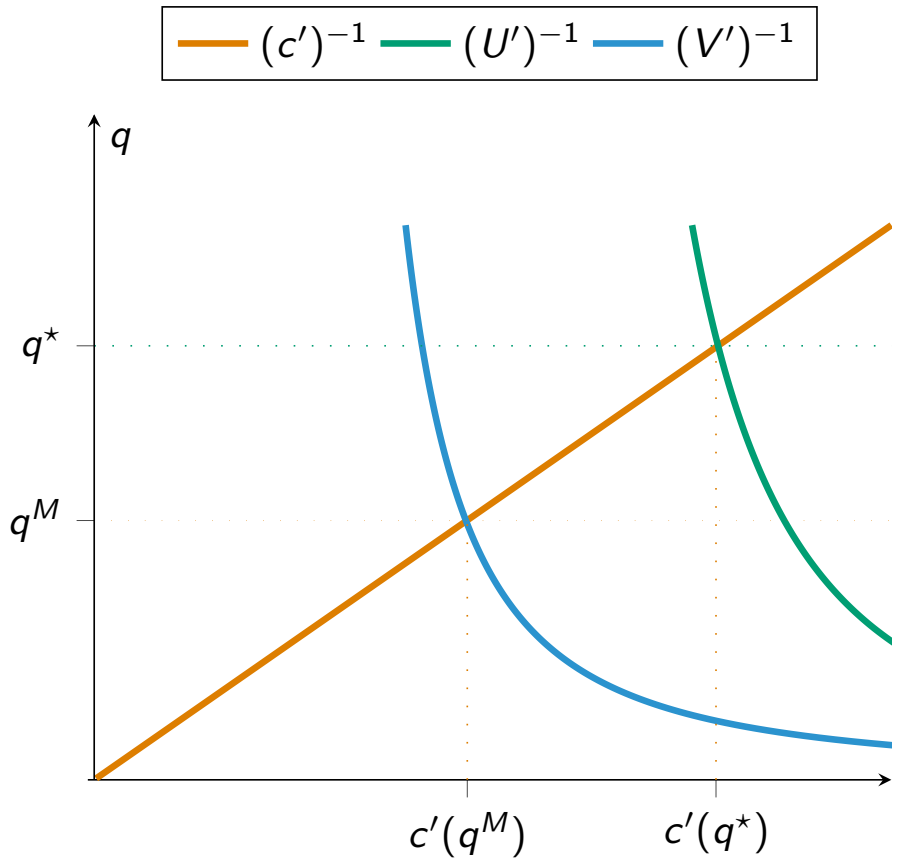


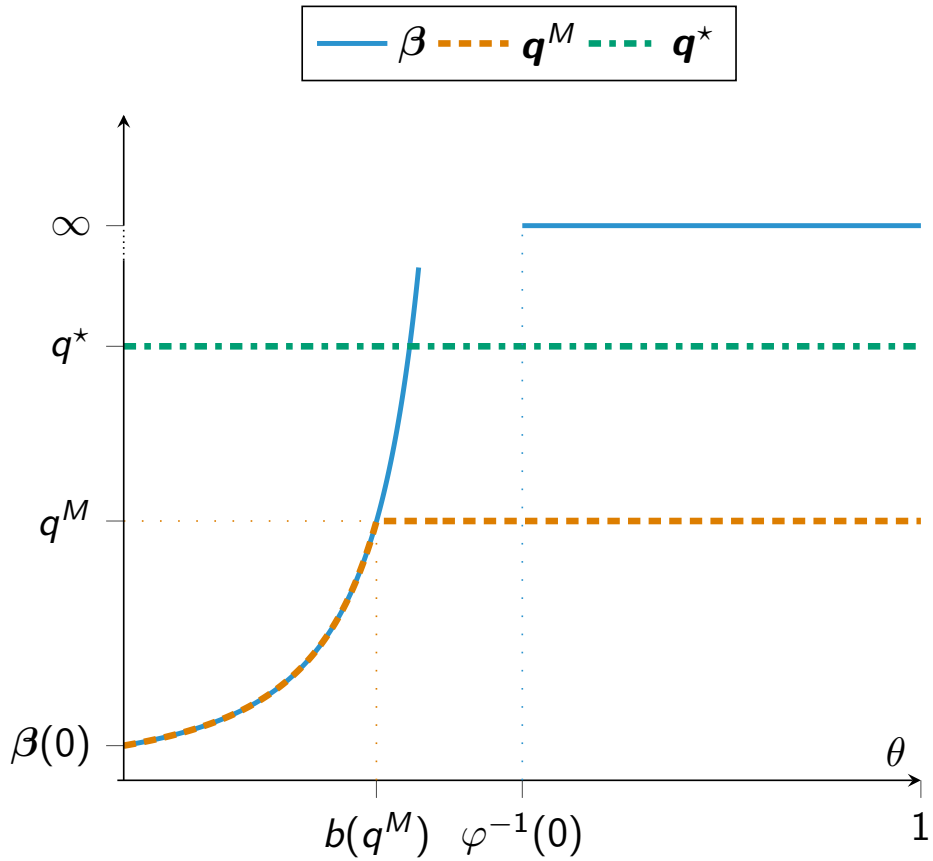


Acquisition

Distribution:
$q^M(\theta) = \min\{\beta(\theta), q^M\}$.

The monopolist allocation





Acquisition

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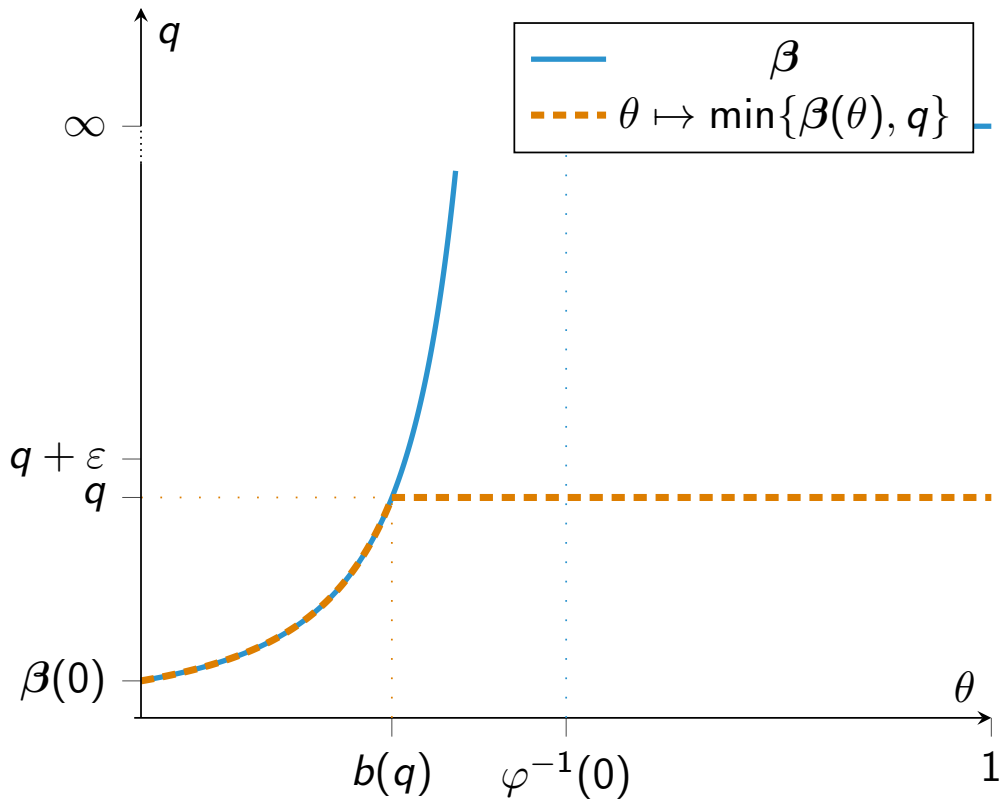
Marginal revenues

The return from increasing the cap of the q-constrained allocation is:

$$V'(q) = \underbrace{(1 - F(b(q)))}_{\text{bunched types}} \underbrace{(g'(q) + b(q))}_{u_q \text{ of } b(q)}.$$

The change from q to $q + \varepsilon$ leads to:

1. same revenues from $q' < q$: q' sold to the same θ, and θ gets the same **rent**;
2. higher quality for bunched types;
3. higher price by $u_q(q, b(q))$.



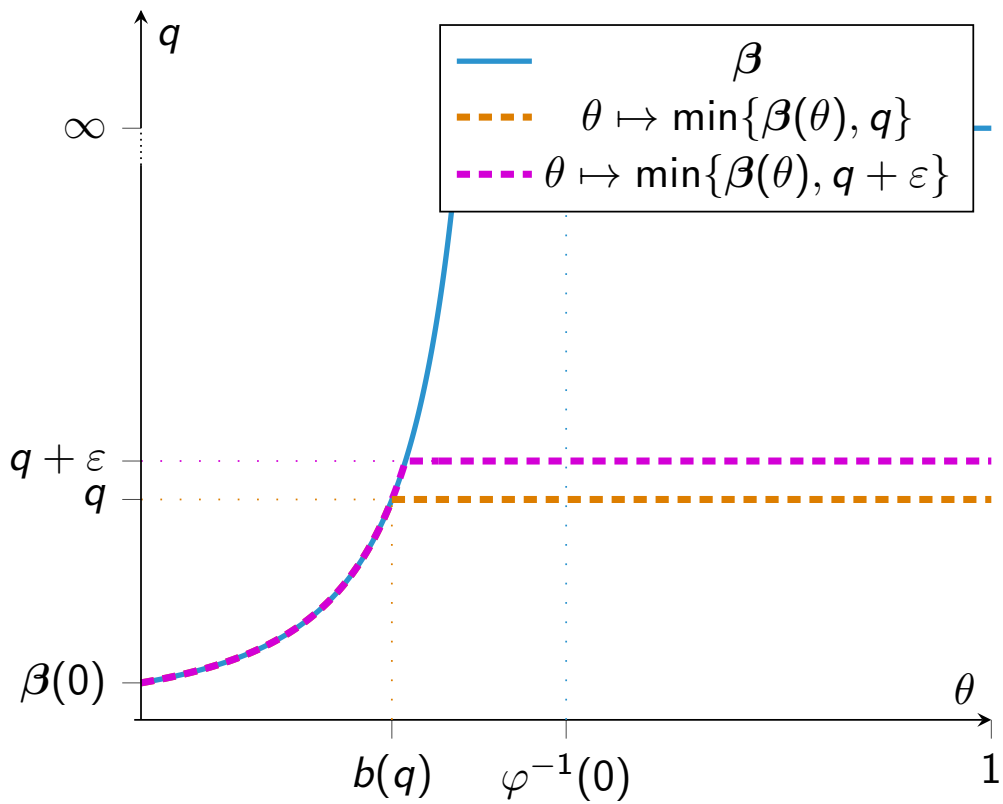
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1. By the distributive properties

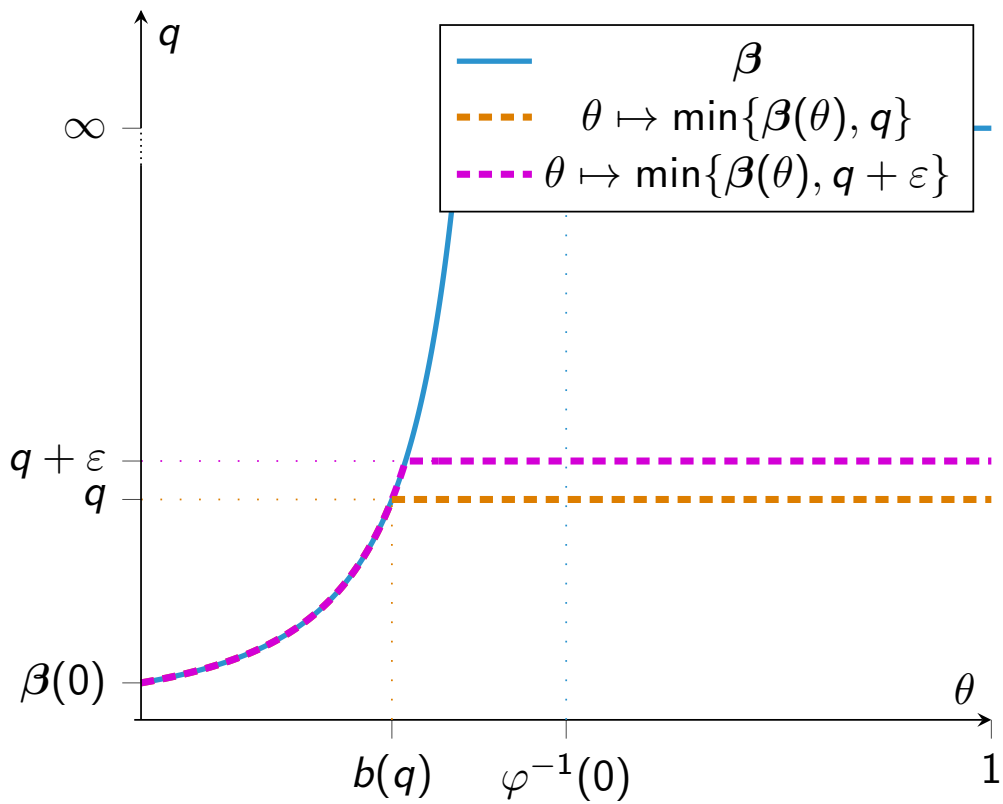
$$1 - F(b(q)) < 1;$$

2. By Markov's inequality:

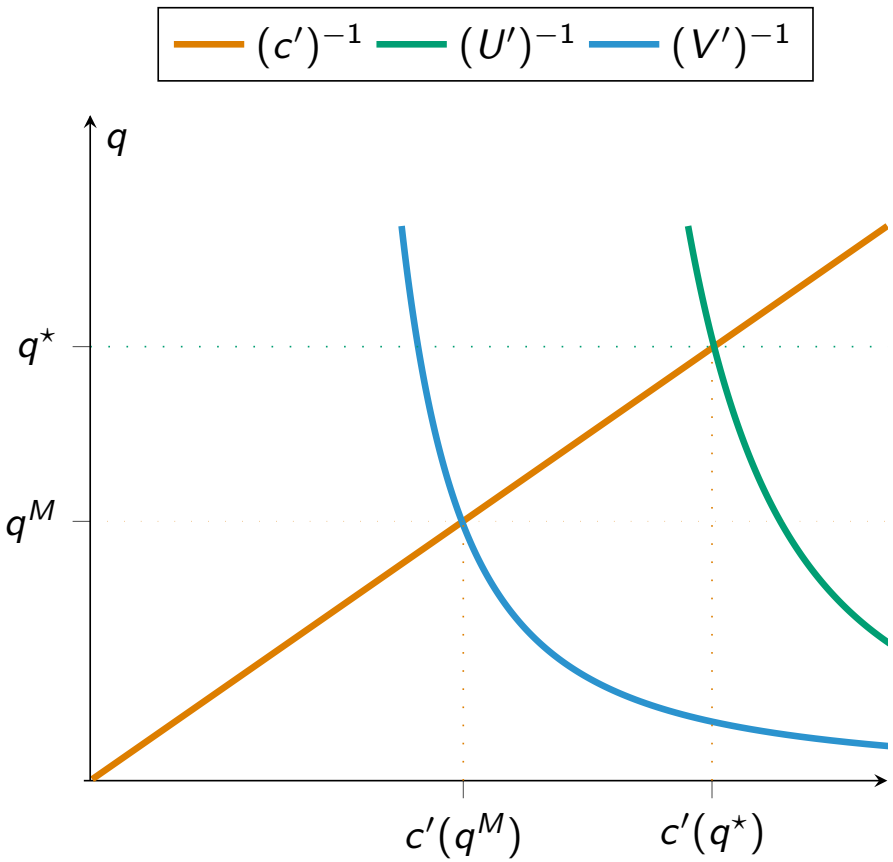
$$(1 - F(b(q)))b(q) \leq \mathbb{E}\{\theta\};$$

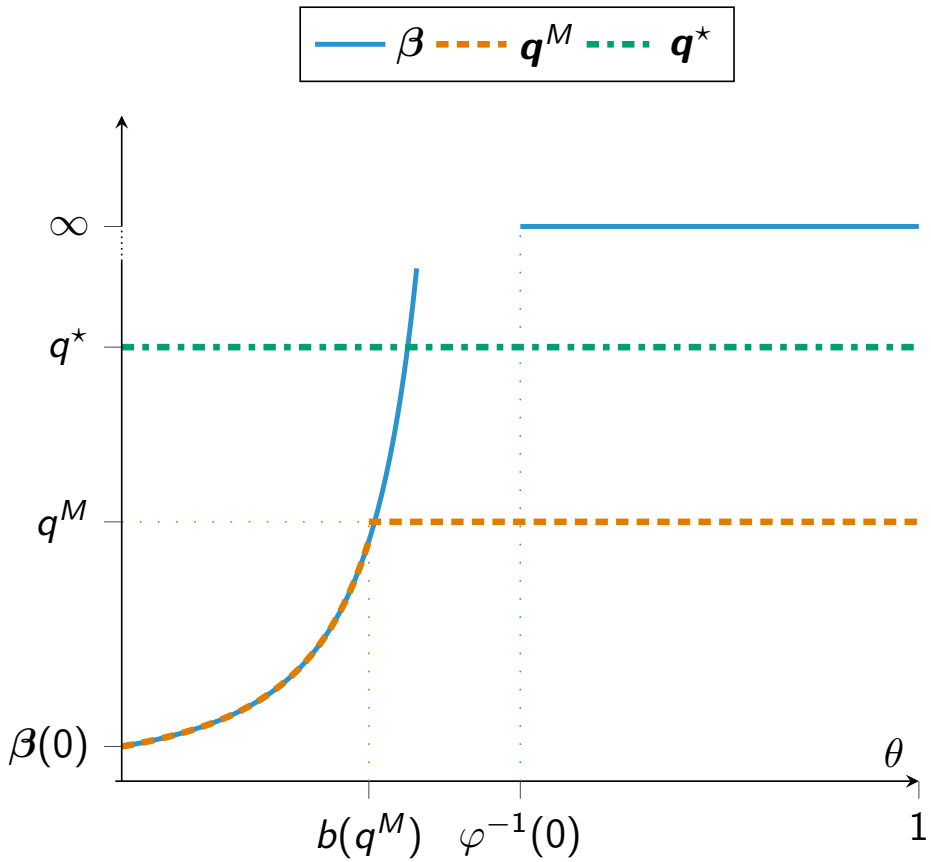
So:

$$V'(q) < g'(q) + \mathbb{E}\{\theta\}.$$



Productive inefficiency





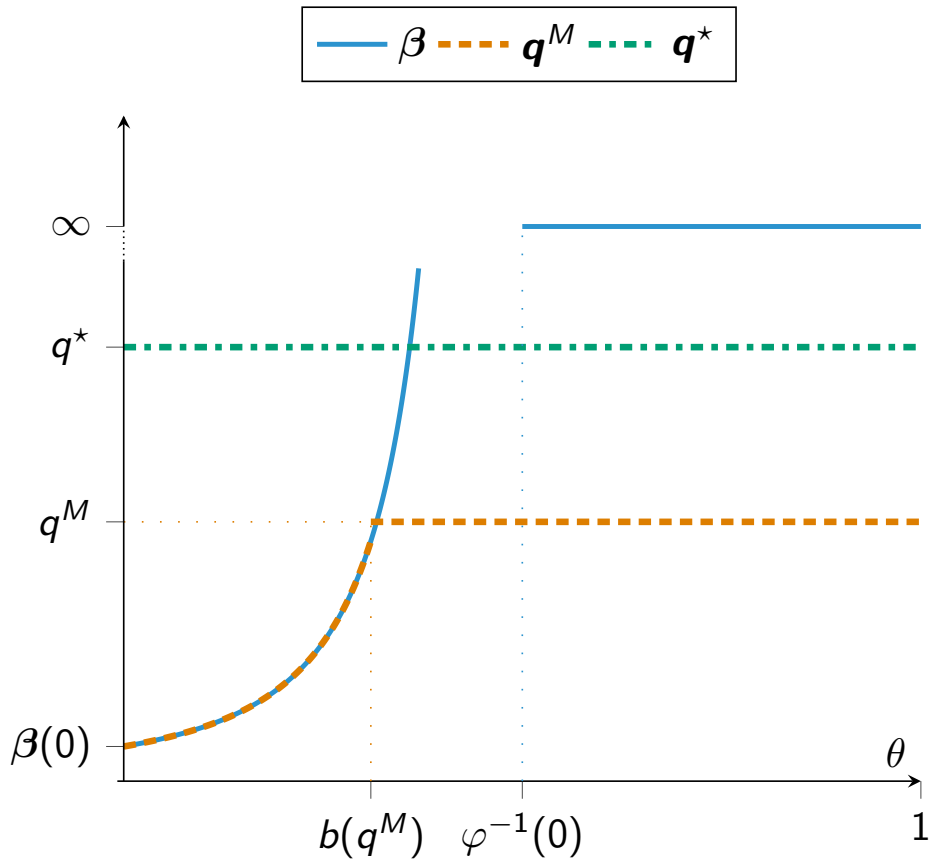
Productive inefficiency

Proposition 2

The monopolist allocation is given by $\boldsymbol{q}^M(\theta) = \min\{\boldsymbol{\beta}(\theta), q\}$ for all θ, in which q^M is the unique q solving

$$V'(q) = c'(q).$$

Moreover, it holds that: $q^M < q^\star$.



No damaging constraint

▸ Competition

No damaging

Without damaging, the q-constrained problem is:

$$V_N(q) := \max_{\boldsymbol{q},\, t(\cdot)} \int_\Theta t(\theta)\, \mathrm{d}F(\theta) \text{ subject to:}$$

$$\text{IC, IR, } \boldsymbol{q}(\theta) \in \{0, q\}, \text{ for all } \theta.$$

The constraint is irrelevant under:
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(Recall: $V'(q) = (1 - F(b(q)))(g'(q) + b(q)), \quad \text{for} \quad g'(q) + \varphi(b(q)) = 0.$)

- Intuitively: damaging ban $\implies$ $n(q) \leq b(q)$, strictly if $b(q) > 0$,
- so productive inefficiency is worse:

$$V'(q) - V_N'(q) = \int_{[n(q), b(q)]} (1 - F(\theta))(g'(q) + \theta)\, \mathrm{d}\theta \geq 0,$$

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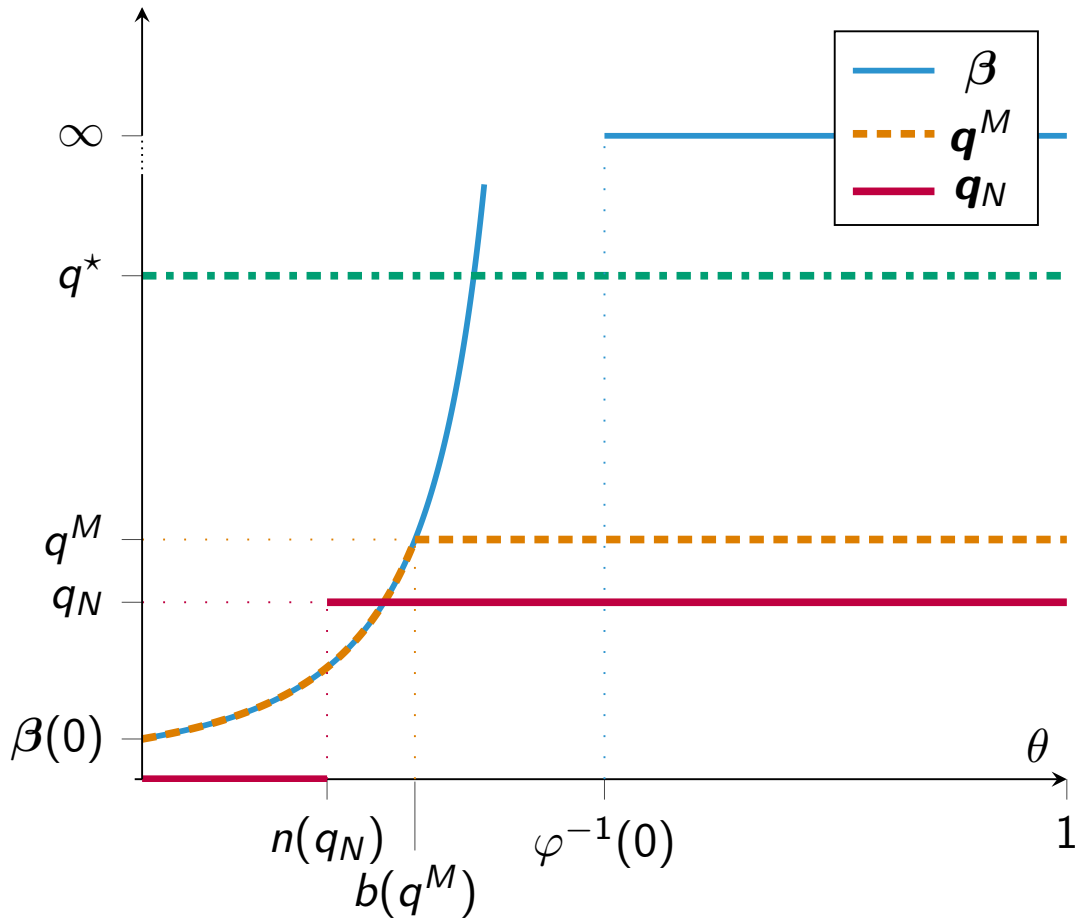
strictly if $b(q) > 0$.

No damaging

The no-damaging allocation $\boldsymbol{q}_N$ features:

- Less production;
- No damaging;
- (Possibility of) exclusion.

The welfare comparison is type specific and ambiguous.

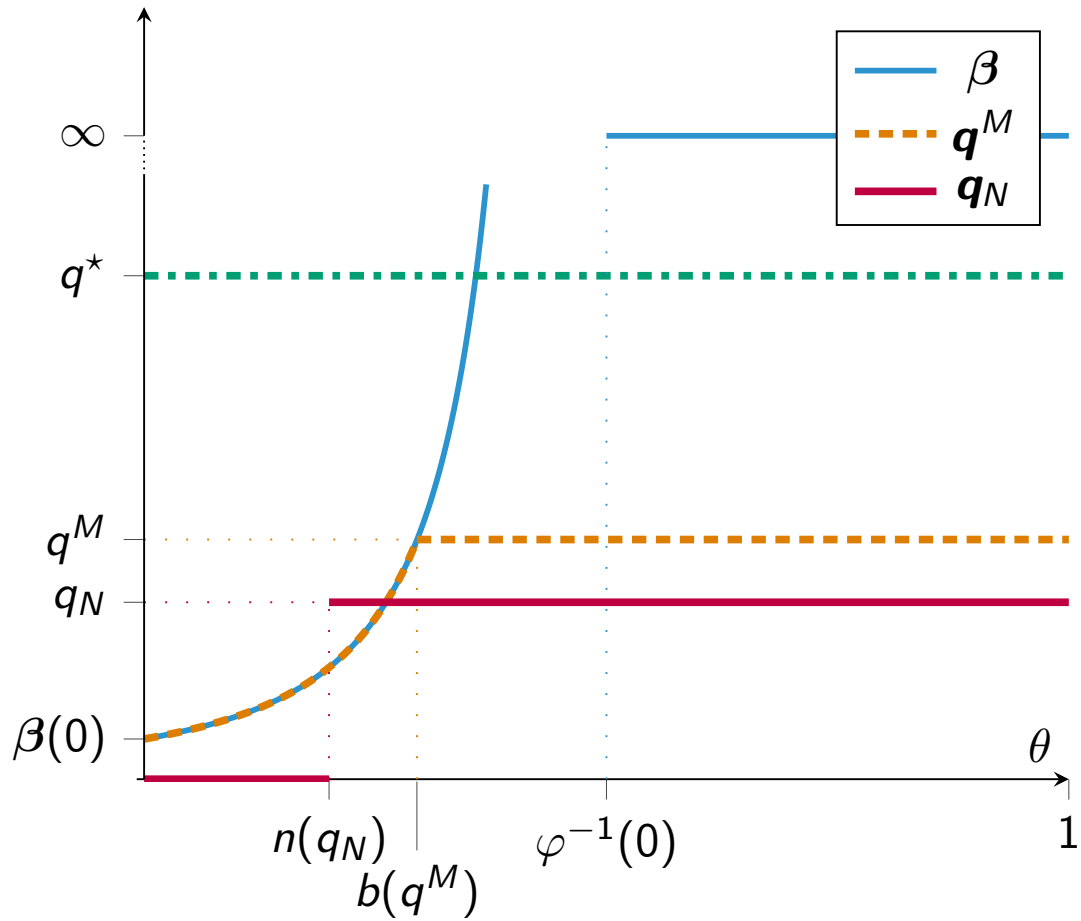


No damaging

The no-damaging allocation $\boldsymbol{q}_N$ features:

- Less production;
- No damaging;
- (Possibility of) exclusion.

The welfare comparison is type specific and ambiguous.



Proposition 3

Without damaging, the monopolist allocation is $\boldsymbol{q}_N(\theta) = \mathbf{1}_{[b_N(q_N),1]}(\theta) q_N$, in which q_N is the unique q solving $V_N'(q) = c'(q)$. Moreover, we have $q_N \leq q^M$, strictly if $b(q^M) > 0$.

Competition

Competition

N firms play the two-stage game of perfect information with timing:

1. Every firm i simultaneously chooses a distribution H_i of her quality, **paying** $c(q_i)$ for the realization q_i;
2. Every firm i simultaneously chooses a pricing function $p_i : \mathbb{R}_+ \to \mathbb{R}$, with $p_i(q) = \infty$ if $q > q_i$, **observing** realized qualities $q_1, \ldots q_n$.

Then, every type θ buys one good from a firm, or does not buy any good for a payoff of 0. (tie breaker favors lowest index.)

Firms play a subgame-perfect Nash equilibrium.

Definition 1

An *n equilibrium* is an equilibrium in which exactly n firms are active.

Second stage

Let's order qualities $(q_1, \ldots, q_N)$ so that: $x \geq y \geq \cdots$

Bertrand force: Every quality below y comes at zero price.

Second stage

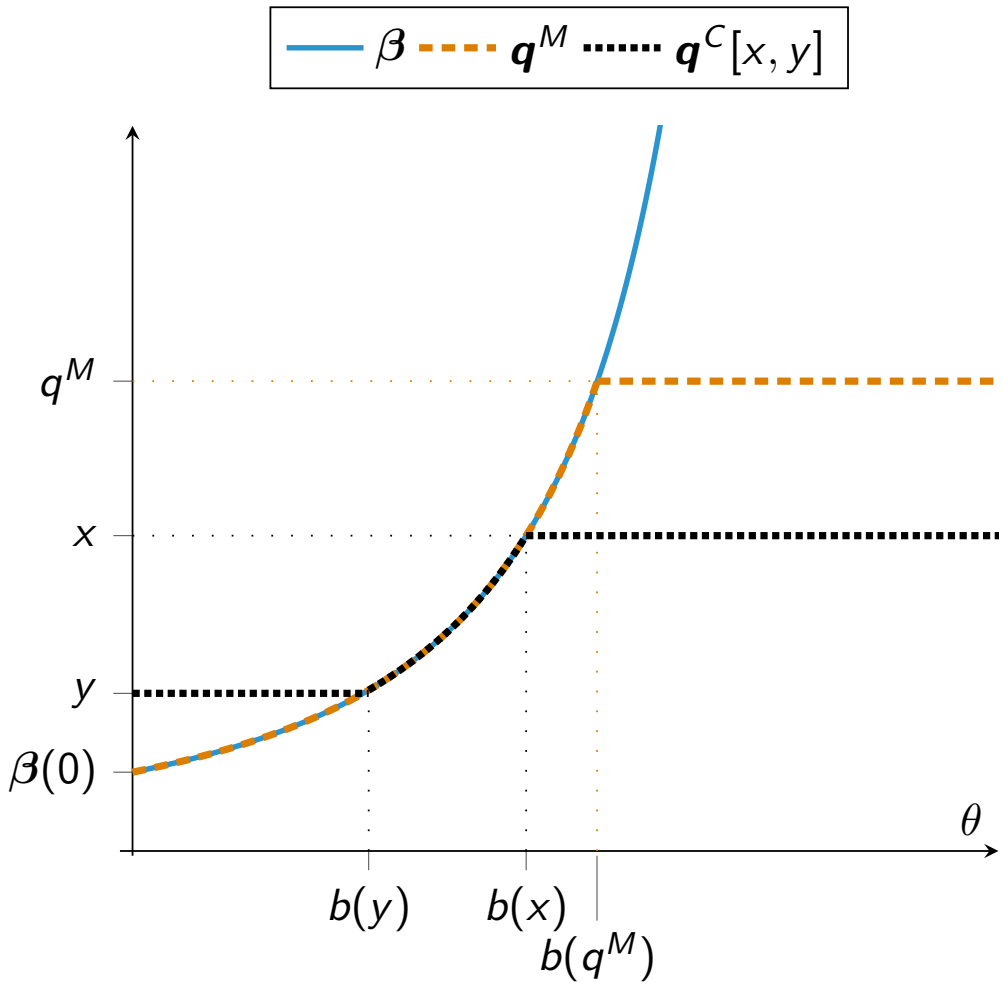
Let's order qualities $(q_1, \ldots, q_N)$ so that: $x \geq y \geq \cdots$

Bertrand force: Every quality below y comes at zero price.

The allocation in the subgame is $\boldsymbol{q}^C[x, y]$, given by:

$$\theta \mapsto \begin{cases} y, & \text{if } \theta \leq b(y), \\ \beta(\theta), & \text{if } \theta \in (b(y), b(x)), \\ x, & \text{if } \theta \geq b(x). \end{cases}$$

Competitive constraint: The "monopolist" earns $V(x) - V(y) - c(x)$

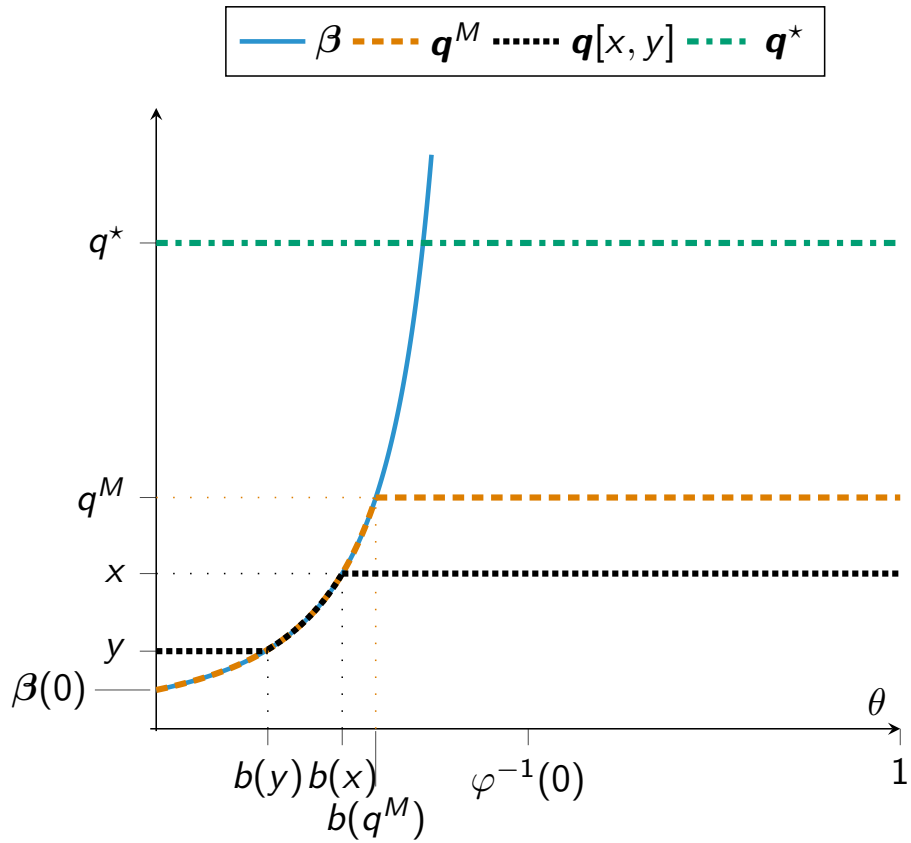


Competitive equilibria

Proposition 4

1. For all $n \leq N$, there exists an n equilibrium.
2. Every competitive n equilibrium induces the random allocation $\boldsymbol{q}^C[\hat{x}, \hat{y}]$, where $\hat{x}$ and $\hat{y}$ are the first and second order statistics of n IID draws from the CDF on $[0, q^M]$ given by

$$H_n(q) = \left(\frac{c'(q)}{V'(q)} \right)^{\frac{1}{n-1}}.$$



Properties of competitive equilibria

- Productive inefficiency is stronger;
- Distributional inefficiency is more prevalent.

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- Productive inefficiency is stronger;
- Distributional inefficiency is more prevalent.

In the paper:

1. Equilibrium welfare with $n \geq 2$ active firms decreases in n.
2. Monopoly dominates duopoly if monopoly fully bunches.
3. Duopoly dominates monopoly if: full bunching does not occur and costs are approximately fixed.

Extra slides

Conclusion

1. With digital goods, two interdependent inefficiencies arise: **productive and damaging**.
2. The *efficiency at the top* insight is revisited: '**distributional efficiency** for **high types**.'

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4. The results extend to: $C(\boldsymbol{q}) = c(\sup \boldsymbol{q}(\Theta)) + \int_{\Theta} k(\boldsymbol{q}(\theta)) \, \mathrm{d}F(\theta)$, increasing-differences u (damaging costs), and nonregular F.
5. **Competition** is beneficial for damaging inefficiency and harmful for productive inefficiency.

▸ More

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▸ More

Thanks!

Monopolistic equilibria

Lemma 3
In every pure-strategy equilibrium: one firm produces q^M and other firms are idle.

Intuition:

1. q^M is the best response to $(0, \ldots, 0)$, and $q \neq q^M$ is **not** a best response to $(q^M, 0, \ldots, 0)$.
2. In any $(x, y, \ldots)$, $y > 0$ is not a best response, and $(0, \ldots, 0)$ is not an equilibrium.

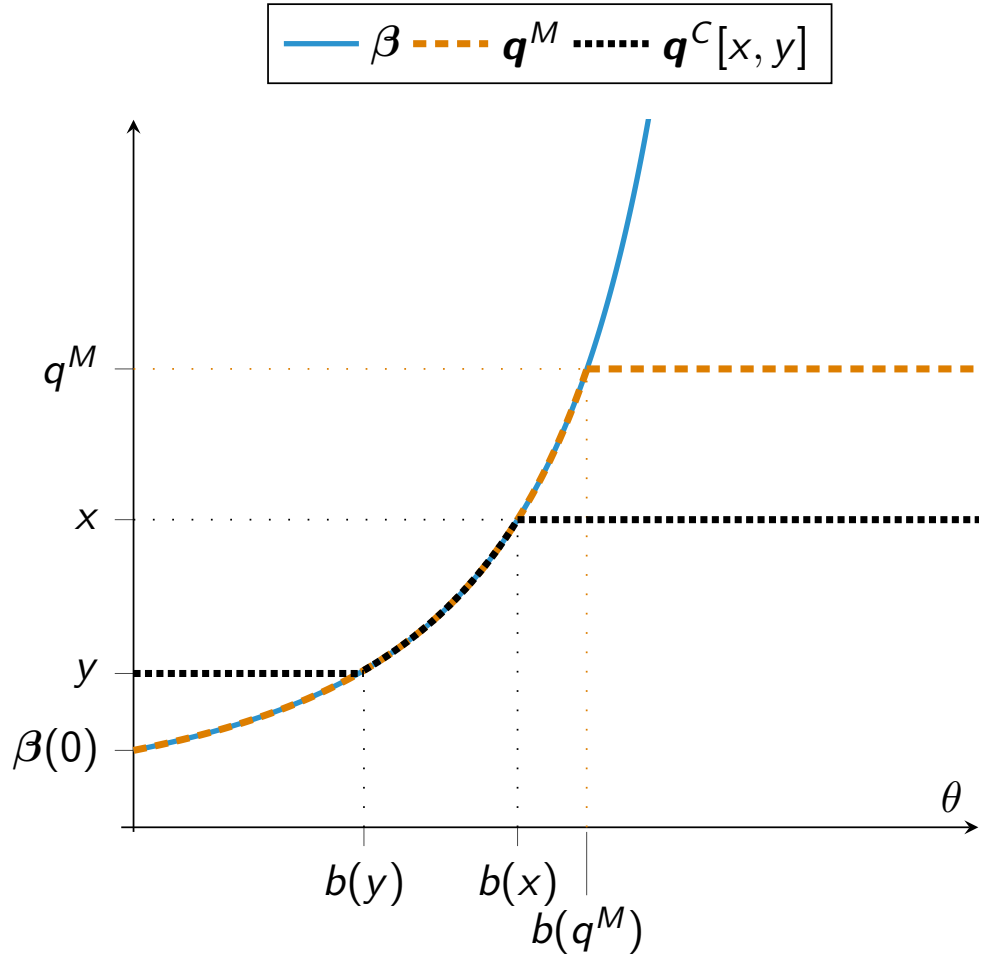
$\Longrightarrow$ Every n equilibrium is mixed if $n \geq 2$ (*competitive*.)

Competitive equilibria

Corollary 1

Every competitive equilibrium leads to an allocation such that, with probability one:

1. The lowest quality is free and $\geq \beta(0)$;
2. The highest quality is strictly lower than q^M;
3. The marginally bunched type is lower than $b(q^M)$.



Literature

Monopolistic screening
Mussa and Rosen (1978); Maskin and Riley (1984); Wilson (1993) . . .
Costs are separable.

Damaged goods
Oren, Smith, and Wilson (1985); Deneckere and McAfee (1996); Grubb (2009); Corrao, Flynn, and Sastry (2023).
Costs are separable, and consumers can damage the good.

Pricing of information with buyer's private information
Bergemann, Bonatti, and Smolin (2018); Bimpikis, Crapis, and Tahbaz-Salehi (2019); Bergemann and Ottaviani (2021); Yang (2022); Bergemann, Cai, Velegkas, and Zhao (2022); Rodríguez Olivera (2024); Bonatti, Dahleh, Horel, and Nouripour (2024) . . .
Information is sold to strategic buyers, without production.

Kolotilin, Mylovanov, Zapechelnyuk, and Li (2017) . . .
Information is allocated without production.

Mechanism & information design
Mensch and Ravid (2025); Thereze (2025).

Literature

Forms of "market power" that guarantee equilibrium existence (Rothschild and Stiglitz (1976), . . . , Stole (2007)):

1. Incumbent-entrant timing (Borenstein and Rose, 1994; Johnson and Myatt, 2003; Gerardi and Shapiro, 2009; Crawford et al., 2019; Boik and Takahashi, 2020);
2. Rich buyer heterogeneity (Rochet and Stole, 2002; Lehmann et al., 2014; Garrett et al., 2019);
3. "Relevance" assumption (Chade and Swinkels, 2021): every firm has a cost advantage over a quality range.

Related models:

1. Kreps and Scheinkman (1983);
   Quantity-and-price competition, no private information, and Cournot outcome;
2. Champsaur and Rochet (1989);
   Quality range-and-price competition, costless quality ranges, which are disjoint in equilibrium.

Separable costs

Cost interpretation

For digital goods: $\Pi(\boldsymbol{q}) := \underbrace{\int_{\Theta} t(\theta)\,\mathrm{d}F(\theta)}_{\text{per-agent revenues}} - \underbrace{c(\sup \boldsymbol{q}(\Theta))}_{\text{per-agent unit cost}}$,

1. Payment $t(\theta)$ and production cost $c(q^M)$ have different size; (Shapiro and Varian, 1998).

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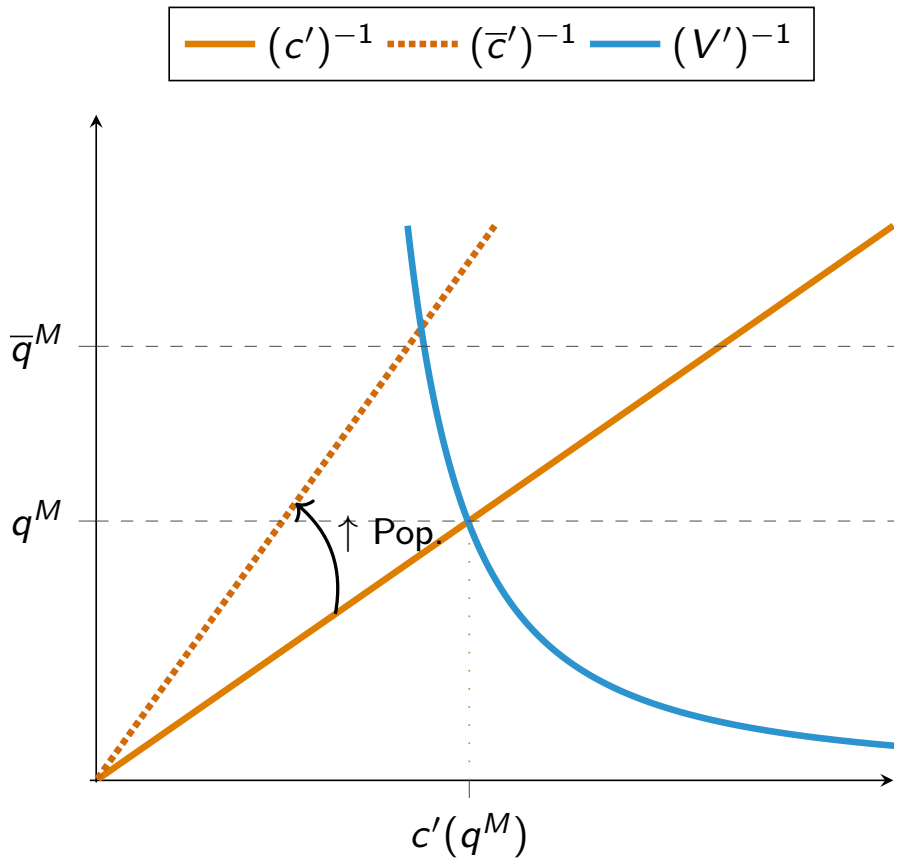
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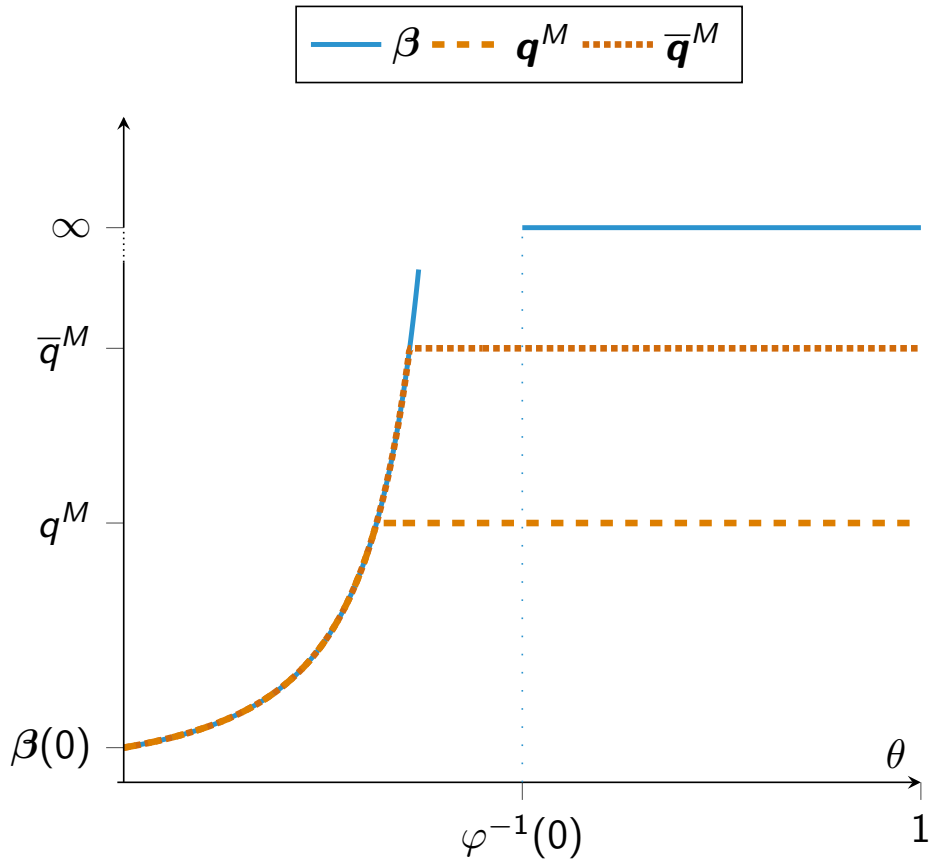
For separable costs: $\Pi^{\text{M-R}}(\boldsymbol{q}) := \underbrace{\int_{\Theta} t(\theta)\,\mathrm{d}F(\theta)}_{\text{per-agent revenues}} - \underbrace{\int_{\Theta} k(\boldsymbol{q}(\theta))\,\mathrm{d}F(\theta)}_{\text{per-agent costs}},$

1. Payment $t(\theta)$ and production cost $k(\boldsymbol{q}(\theta))$ are comparable;
2. Population size only scales profits;

In general: $C(\boldsymbol{q}) = \int_{\Theta} k(\boldsymbol{q}(\theta))\,\mathrm{d}F(\theta) + c(\sup \boldsymbol{q}(\Theta))$.

Population size





Single buyer

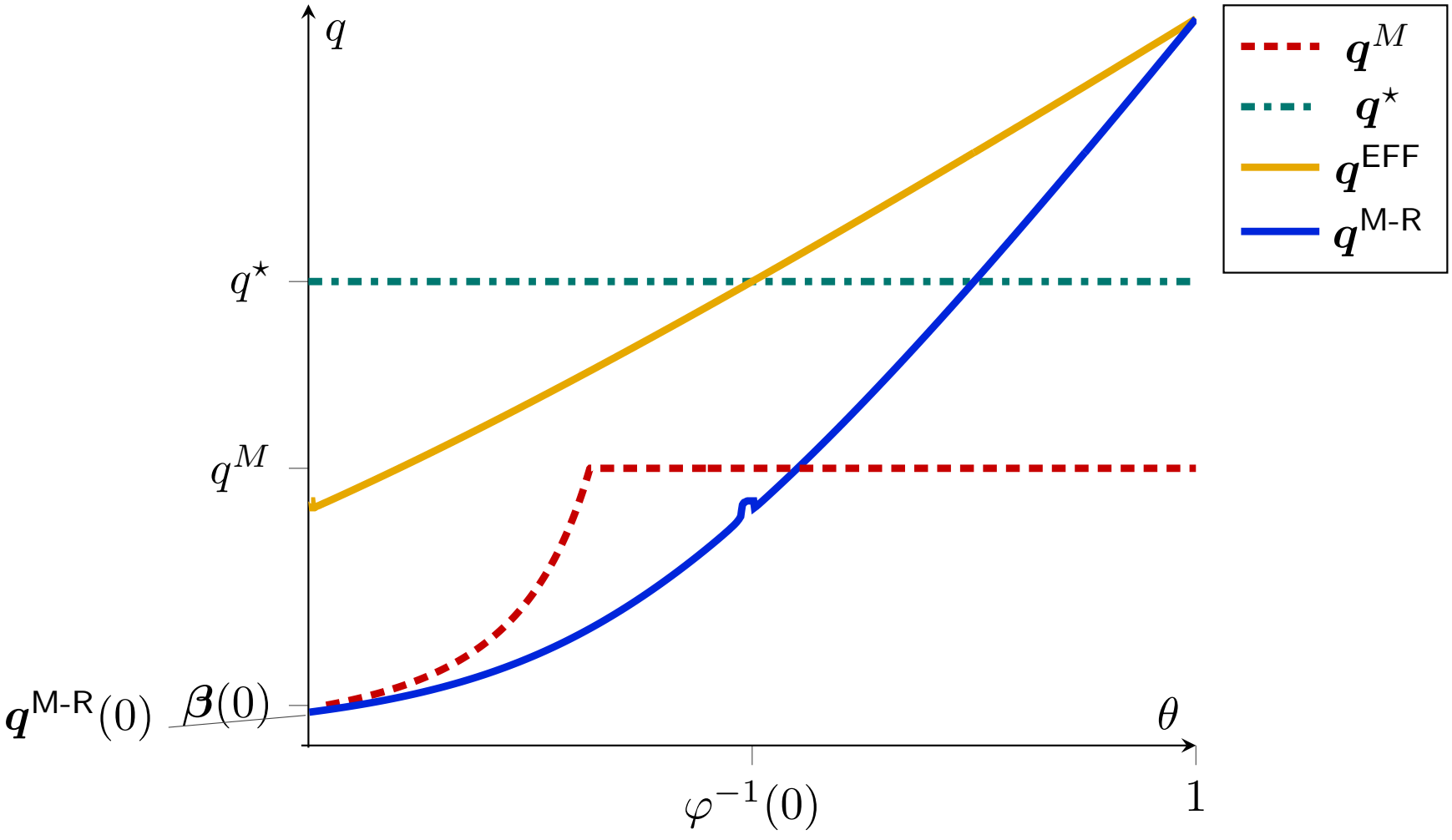
$$\Pi^{\text{M-R}}(\boldsymbol{q}) = \underbrace{\int_{\Theta} t(\theta)\,\mathrm{d}F(\theta)}_{\text{expected revenues}} - \underbrace{\int_{\Theta} c(\boldsymbol{q}(\theta))\,\mathrm{d}F(\theta)}_{\text{expected costs}},$$

1. Payment $t(\theta)$ and production cost $c(\boldsymbol{q}(\theta))$ are comparable;
2. Production occurs **after** eliciting the buyer's type;
3. Free damaging and replication are irrelevant.

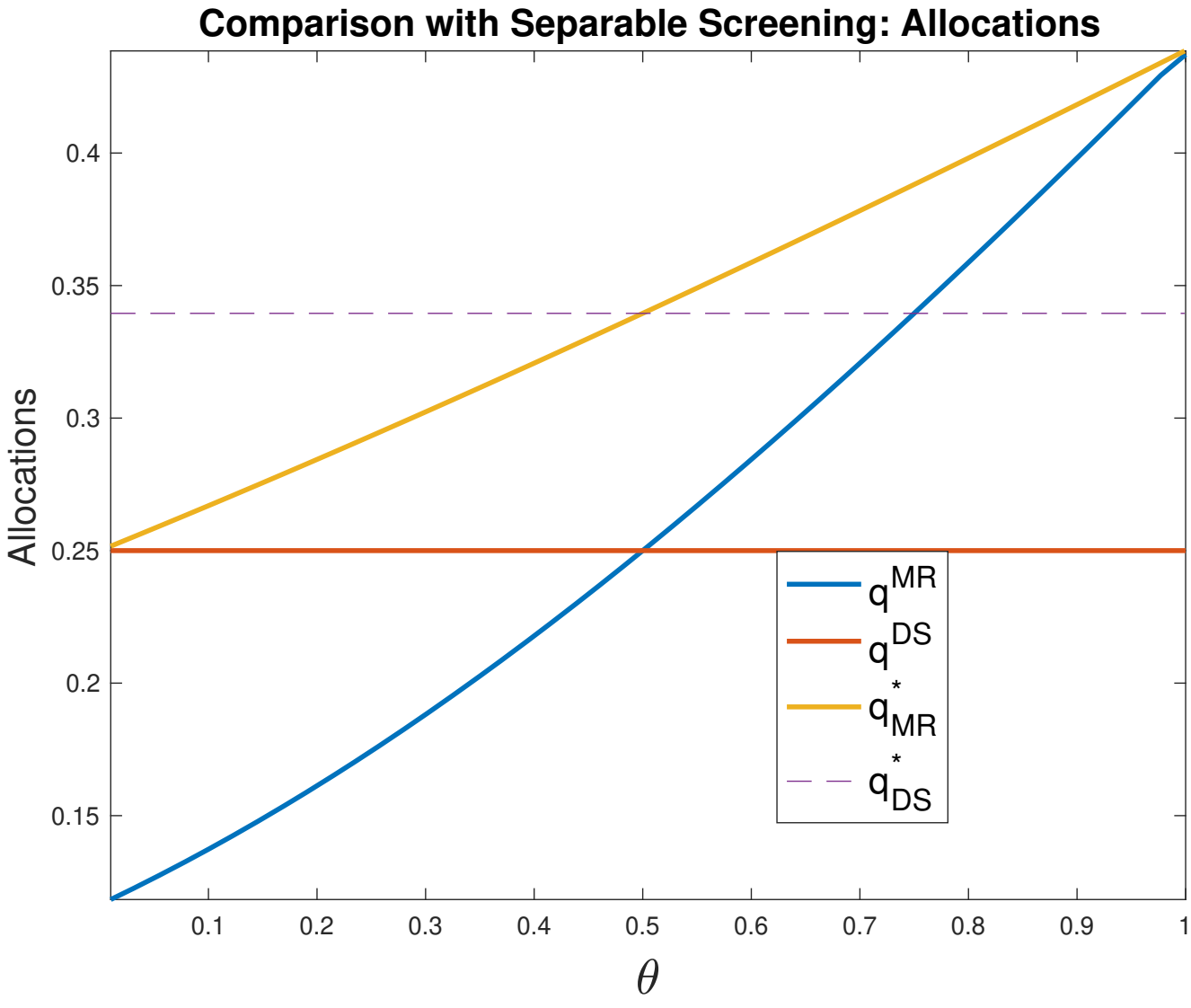
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1. Payment $t(\theta)$ and production cost $c(\sup \boldsymbol{q}(\Theta))$ are comparable;
2. Production occurs **before** eliciting the buyer's type;
3. Free damaging matters, replication is irrelevant.

Single buyer



Single buyer



More

Hybrid costs

With more general costs: $C(\boldsymbol{q}) = \int_\Theta \hat{c}(\boldsymbol{q}(\theta), \sup \boldsymbol{q}(\Theta))\, \mathrm{d}F(\theta)$,
the seller pays:

1. Development / production costs: $\sup \boldsymbol{q}(\Theta)$;
2. Distribution / replication / damaging costs: $\boldsymbol{q}(\theta)$.

Lemma 1 holds, but the characterization of $\boldsymbol{q}^M$ has two complications:

1. Distribution: the solution to $\mathcal{P}(q)$ does not depends on q solely through capping;
2. Production: the marginal return $V'(q)$ depends on: (i) bunching region, and (ii) damaging.

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1. Distribution: the solution to $\mathcal{P}(q)$ does not depends on q solely through capping;
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If $C(\boldsymbol{q}) = c(\sup \boldsymbol{q}(\Theta)) + \kappa \log\left(\frac{\sup \boldsymbol{q}(\Theta)}{\boldsymbol{q}(\theta)}\right)$, then **1.** is turned off.

Damaging costs

If $C(\boldsymbol{q}) = c(\sup \boldsymbol{q}(\Theta)) + \kappa \log\left(\frac{\sup \boldsymbol{q}(\Theta)}{\boldsymbol{q}(\theta)}\right)$, then:

- Production costs + pure-damaging replication / distribution costs;

Damaging costs

If $C(\boldsymbol{q}) = c(\sup \boldsymbol{q}(\Theta)) + \kappa \log\left(\frac{\sup \boldsymbol{q}(\Theta)}{\boldsymbol{q}(\theta)}\right)$, then:

- Production costs + pure-damaging replication / distribution costs;
- The efficient allocation is flat: damaging decreases utility and increases costs;
- The solution to $\mathcal{P}(q)$ is $\theta \mapsto \min\{\boldsymbol{\beta}_\kappa(\theta), q\}$.

1. $\kappa > 0$ acts as a preference shift ($\uparrow g$) at the distribution stage:
   - $\uparrow \boldsymbol{\beta}_\kappa$ and $\downarrow b_\kappa(q)$;

Damaging costs

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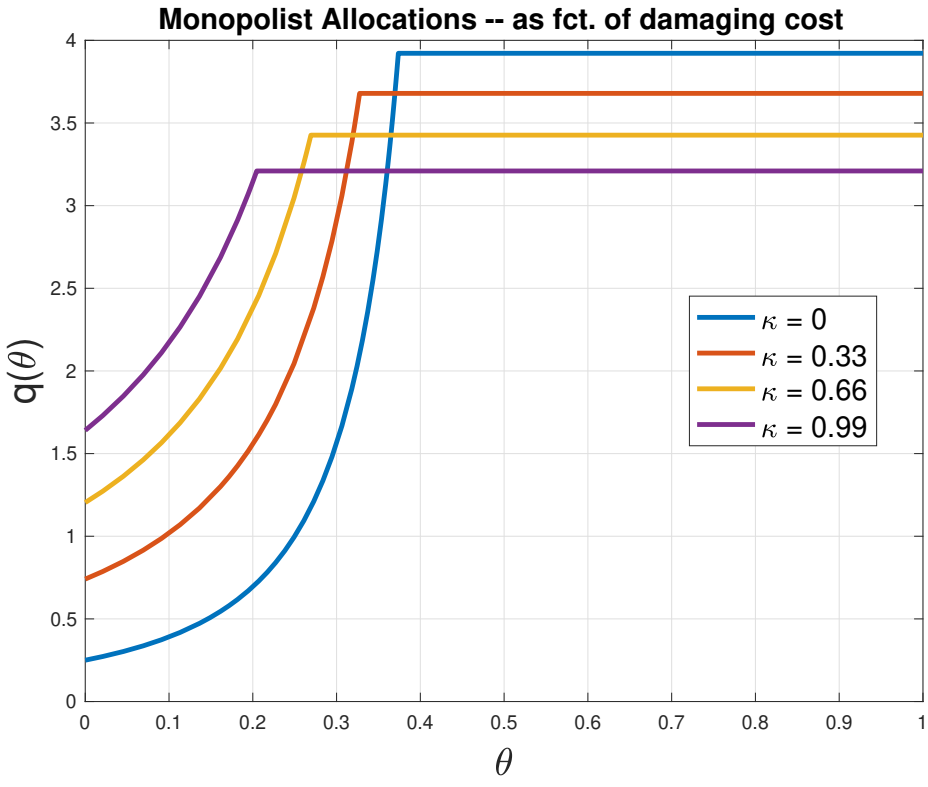
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1. $\kappa > 0$ acts as a preference shift ($\uparrow g$) at the distribution stage:
   - $\uparrow \boldsymbol{\beta}_\kappa$ and $\downarrow b_\kappa(q)$;
2. $\kappa > 0$ impacts production directly:
   - $V'(q) = (1 - F(b_\kappa(q))(g'(q) + b_\kappa(q)) - \kappa \frac{b_\kappa(q)}{q}$.

Damaging costs

$\kappa > 0$ implies

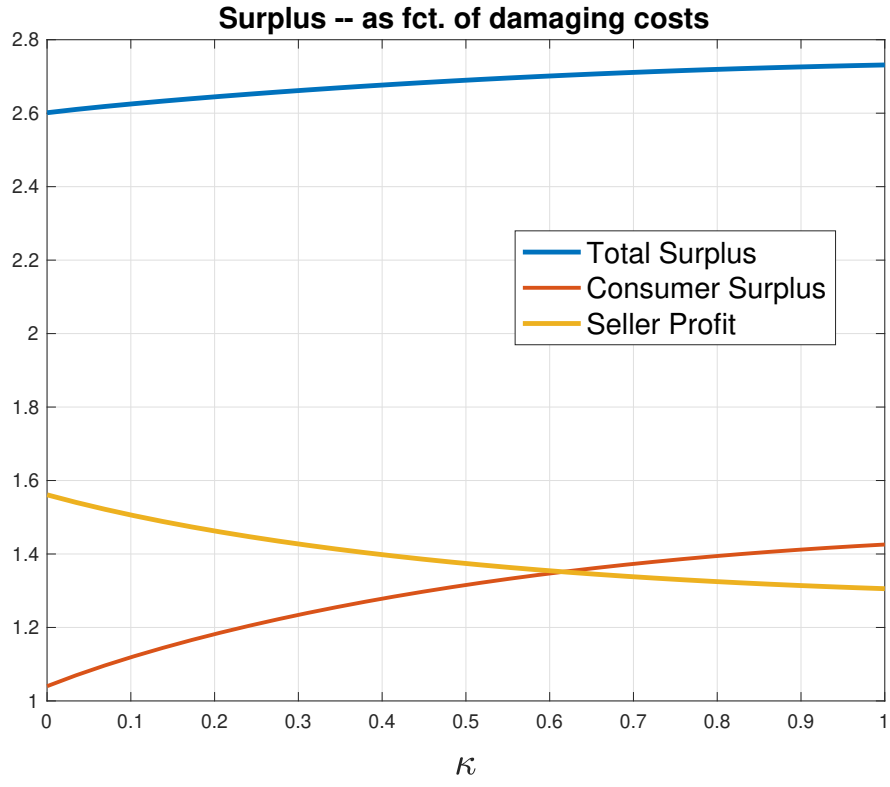
1. Less damaging;
2. Lower production.



Damaging costs

$\kappa > 0$ implies

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Efficiency with general u and k

Proposition 5

The allocation $\boldsymbol{q}^\star$ is efficient iff $\boldsymbol{q}^\star(\theta) = \min\{\gamma(\theta), q^\star\}$ for all θ, in which: $q^\star$ is the unique q such that $\int_{[a(q),1]} u_1(q,\theta) - k'(q)\,\mathrm{d}F(\theta) = c'(q)$, and γ is an allocation such that $\gamma(\theta) = \boldsymbol{\alpha}(\theta)$ almost everywhere.

In general, $q \in [0, \overline{q}]$, and:

$$J(q,\theta) := u(q,\theta) - \frac{1}{h(\theta)} u_2(q,\theta) - k(q),$$

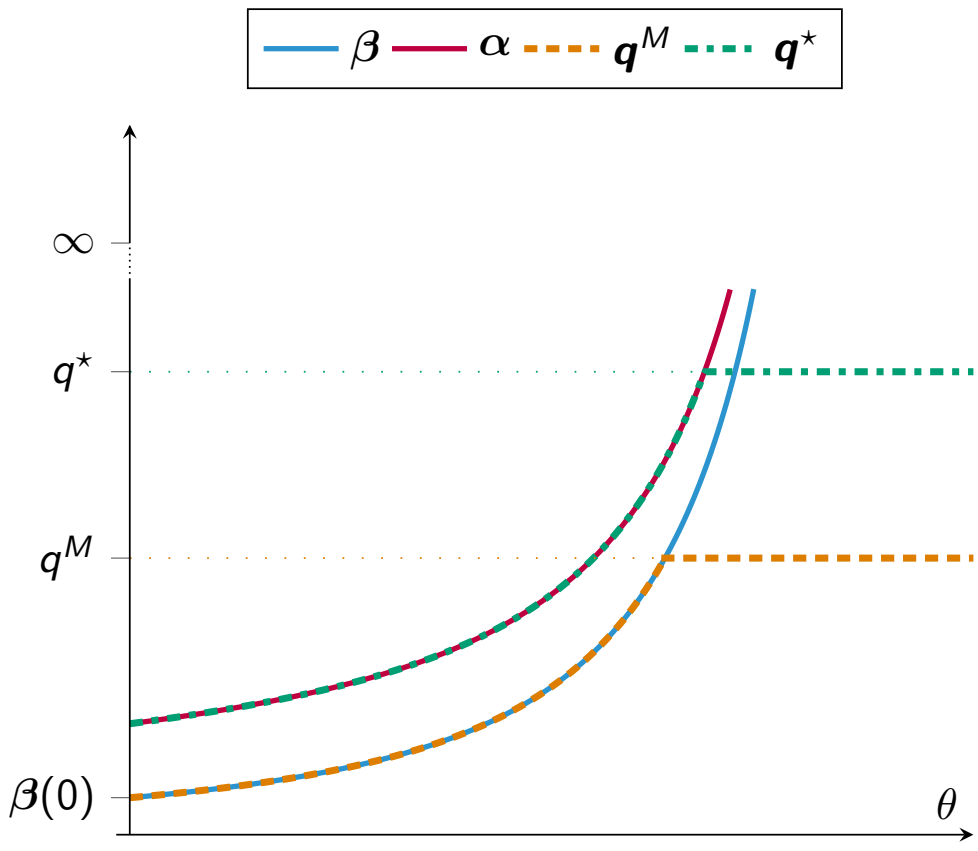
$\boldsymbol{\beta}(\theta)$ is the largest element of

$$\operatorname*{Argmax}_q J(q,\theta),$$

$\boldsymbol{\alpha}(\theta)$ is the largest element of

$$\operatorname*{Argmax}_q u(q,\theta) - k(q),$$

u and J satisfy incr. differences, and are: twice diff., concave in q for all θ, str. quasiconcave in q a.e. on Θ; k is Inada.



Monopoly with general u and k

Proposition 6

The allocation $\boldsymbol{q}^M$ is monopolist iff $\boldsymbol{q}^M(\theta) = \min\{\gamma(\theta), q^M\}$ for all θ, in which: q^M is the unique q such that $\int_{[b(q),1]} J_1(q,\theta)\,\mathrm{d}F(\theta) = c'(q)$, and γ is a nondecreasing allocation such that $\gamma(\theta) = \boldsymbol{\beta}(\theta)$ almost everywhere. Moreover, $0 < q^M < q^\star$.

In general, $q \in [0, \overline{q}]$, and:

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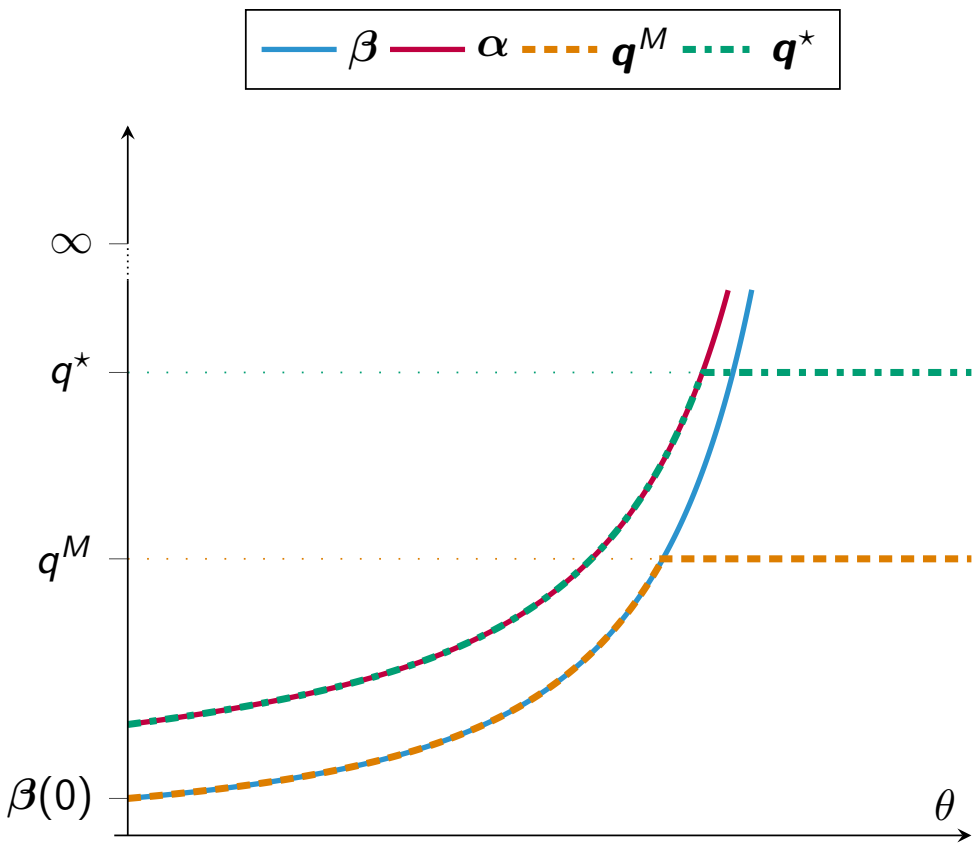
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Back

No-damaging monopoly with general u and k

Assumption: $J(0,\theta) = 0$ for all θ and $J(q,\cdot)$ is increasing for all $q > 0$.

Proposition 7

The allocation $\boldsymbol{q}_N$ is no screening iff $\boldsymbol{q}_N(\theta) = [\theta \geq b_N(q_N)]q_N$ for all $\theta \neq b_N(q_N)$ and $\boldsymbol{q}_N(b_N(q_N)) \in \{0, q_N\}$, in which q_N is the unique q such that: $\int_{[b_N(q),1]} J_1(q,\theta)\,\mathrm{d}F(\theta) = c'(q)$. Moreover, it holds that:

1. $0 < q_N \leq q^M$;
2. $q_N < q^M$ if $b(q^M) > b_N(q^M)$.

We use Iverson brackets: $[P] = 1$ if P is true, and $[P] = 0$ otherwise.

Productive inefficiency addendum 1/3

Productive inefficiency arises if:

$$\underbrace{V'(q)}_{\substack{\text{Marginal revenues} \\ \text{given } \theta \mapsto \min\{\beta(\theta), q\}}} < \underbrace{g'(q) + \mathbb{E}\{\theta\}}_{\substack{\text{Marginal total utility} \\ \text{given } \theta \mapsto q}} .$$

1. The q constrained allocation $\theta \mapsto \min\{\beta(\theta), q\}$ induces total utility

$$U(q) := \mathbb{E}\{u(\min\{\beta(\theta), q\}, \theta)\};$$

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2. $U'(q) <$ marginal total utility given $\theta \mapsto q$, because

$$U'(q) = \int_{[b(q),1]} g'(q) + \theta \, \mathrm{d}F(\theta) \leq g'(q) + \mathbb{E}\{\theta\};$$

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$$V'(q) < V'(q) + \text{Rents}'(q) = U'(q).$$

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Productive inefficiency addendum 2/3

Productive inefficiency arises because:

$$V'(q) < g'(q) + \mathbb{E}\{\theta\}.$$

The monopoly that produces quality q implies total surplus

$$V(q) + U(q) - c(q),$$

with $U(q) = \int_{[0,1]} \int_{[0,\theta]} \min\{\beta(\theta'), \overline{q}\} \, \mathrm{d}\theta' \, \mathrm{d}F(\theta)$ (Envelope Theorem).

The marginal surplus is $V'(q) + U'(q)$ and satisfies

$$V'(q) < V'(q) + U'(q) \leq g'(q) + \mathbb{E}\{\theta\}.$$

1. Monopolist does not internalize buyer surplus;
2. Damaging inefficiency.

Productive inefficiency addendum 3/3

WTS: $V'(q) + U'(q) \leq g'(q) + \mathbb{E}\{\theta\}$.

1. $U'(q) = \int_{[b(q),1]} \theta - b(q)\,\mathrm{d}F(\theta)$,
   because the marginal $u(\boldsymbol{q}(\theta), \theta)$ increases at rate $g'(q) + \theta$ and the marginal transfer at rate $g'(q) + b(q)$, for $\theta > b(q)$ and $\boldsymbol{q}(\cdot) = \min\{\beta(\cdot), q\}$;
2. Using $V'(q) = (1 - F(b(q)))(g'(q) + b(q))$, we have

$$V'(q) + U'(q) = (1 - F(b(q)))g'(q) + \int_{[b(q),1]} \theta\,\mathrm{d}F(\theta).$$

Note that $U'(q) > 0$ for all $q > 0$, because $b(q) \leq \varphi^{-1}(0) < 1$ for all $q \geq 0$.)

The game

Type θ buys quality $D_p(\theta) \in Q := \mathbb{R}_+$ from firm $\iota_p(\theta)$, given the pricing functions in $(p_1, \dots, p_N) = p$. The set of firms is $\mathcal{N} = (1, \dots, N)$.

The revenues of i given the pricing functions in $(p_1, \dots, p_N) = p$ are

$$R_i(p_1, \dots, p_N) := \int_{\{\theta | \iota_p(\theta) = i\}} p_i(D_p(\theta)) \, \mathrm{d}F(\theta).$$

The set of pure strategies for firm i is $S_i := Q \times \boldsymbol{P}_i$, letting $\boldsymbol{P}_i \subseteq (\mathbb{R}^Q)^{Q^N}$ be the set of "conditional" pricing functions of firm i.

The *payoff* of firm i from the profile $s := (\dots, (\overline{q}_i^s, P_i^s), \dots) \in \times_{i=1}^N S_i$ is

$$\Pi_i(s) := R_i(P_1^s[\overline{q}^s], \dots, P_N^s[\overline{q}^s]) - c(\overline{q}_i^s).$$

The equilibrium

A strategy for player i is a pair of a cap distribution and conditional pricing function, (H_i, P_i), in which H_i is a distribution function with support contained in Q and $P_i \in \mathbf{P}_i$ for all i.

A strategy profile $((H_1, P_1), \ldots, (H_N, P_N))$ is an *equilibrium* if it is a subgame-perfect Nash equilibrium of the game:

1. The conditional pricing functions maximize conditional profits, that is, for all cap profiles $\overline{q} = (\overline{q}_1, \ldots, \overline{q}_N)$,

$$R_i(P_i[\overline{q}], P_{-i}[\overline{q}]) \geq R_i(p_i', P_{-i}[\overline{q}]) \quad \text{for all } p_i' \in \mathbb{R}^Q, \ i.$$

2. The cap distributions maximize expected profits, that is, for all $\overline{q}_i$ in the support of H_i we have

$$\int_{Q^{N-1}} R_i(P_i[\overline{q}], P_{-i}[\overline{q}]) \, \mathrm{d}H_{-i}(\overline{q}_{-i}) - c(\overline{q}_i) \geq$$

$$\int_{Q^{N-1}} R_i(P_i[\overline{q}_i', \overline{q}_{-i}], P_{-i}[\overline{q}_i', \overline{q}_{-i}]) \, \mathrm{d}H_{-i}(\overline{q}_{-i}) - c(\overline{q}_i')$$

for all $\overline{q}_i' \in Q$, letting H_{-i} denote the joint distribution of the caps of i's opponents under $((H_1, P_1), \ldots, (H_N, P_N))$, and for all i.

Auxiliary monopoly problem

The allocation $\boldsymbol{q}$ is x-y *second best*, for qualities x, y with $x \leq y$, if there exists a transfer function $t(\cdot)$ such that: $(\boldsymbol{q}, t(\cdot))$ solves

$$\sup_{\boldsymbol{q},\, t(\cdot)} \int t(\theta)\, \mathrm{d}F(\theta) \text{ subject to: } y \leq \boldsymbol{q}(\theta) \leq x \text{ for all } \theta \in \Theta.$$

subject to: IC, IR.

Lemma 4

The allocation $\boldsymbol{q}$ is x-y second best iff $\boldsymbol{q}(\theta) = \max\{\min\{\boldsymbol{\gamma}(\theta), x\}, y\}$ for all θ, in which $\boldsymbol{\gamma}$ is a nondecreasing allocation such that $\boldsymbol{\gamma}(\theta) = \boldsymbol{\beta}(\theta)$ almost everywhere.

The pricing subgame

The pricing game given the quality profile $(q_1, \ldots, q_N)$ is the strategic-form game $\Gamma(q_1, \ldots, q_N) = \left(\mathcal{N}, (\mathbb{R}^{[0,q_i]}, R_i(\cdot))_{i\in\mathcal{N}}\right)$.

The profile of pricing functions $(p_1, \ldots, p_N)$ is a $(q_1, \ldots, q_N)$ *equilibrium* if $(p_1, \ldots, p_N)$ is a Nash equilibrium of $\Gamma(q_1, \ldots, q_N)$.

We study the subgame starting at the given production profile $(q_1, \ldots, q_N) \in Q^N$ with $x := \max\{q_1, \ldots, q_N\}x$ and $y := \max\{q_1, \ldots, q_N\} \setminus \{x\}$.

Lemma 5

For every $(q_1, \ldots, q_N)$ equilibrium $(p_1, \ldots, p_N)$, the following formulas hold:

$$R_i(p_1, \ldots, p_N) = (V(q_i) - V(y))_+,$$

and

$$D_{(p_1,\ldots,p_N)}(\theta) = \max\{\min\{\gamma(\theta), x\}, y\},$$

for all θ, in which γ is a nondecreasing allocation such that $\gamma(\theta) = \beta(\theta)$ almost everywhere.

The production game

Every Nash equilibrium of $\Gamma(q_1, \ldots, q_N)$ induces the same revenues, i.e.,

$$R_i(p_1^*, \ldots, p_N^*) = R_i(p_1^{**}, \ldots, p_N^{**})$$

for all $(q_1, \ldots, q_N)$ equilibria $(p_1^*, \ldots, p_N^*)$, $(p_1^{**}, \ldots, p_N^{**})$. We call $\overline{R}_i(q_1, \ldots, q_N)$ the unique equilibrium revenues of i in the pricing game $\Gamma(q_1, \ldots, q_N)$, that is:

$$\overline{R}_i(q_1, \ldots, q_N) = (V(q_i) - V(y))_+.$$

The *production game* is the strategic-form game $\Gamma = \left(\mathcal{N}, (Q, \overline{R}_i(\cdot) - c(\cdot))_{i \in \mathcal{N}}\right)$. By mixed strategies, we refer to probability distributions over Q identified by ther distribution functions, and we extend payoffs of Γ to mixed-strategy profiles defining

$$\Pi_i \colon (\Delta Q)^N \to \mathbb{R}$$

$$(H_1, \ldots, H_N) \mapsto \int_Q \cdots \int_Q \overline{R}_i(q_1, \ldots, q_N) - c(q_i)\, \mathrm{d}H_1(q_1) \cdots \mathrm{d}H_N(q_N).$$

Auxiliary result for the production game

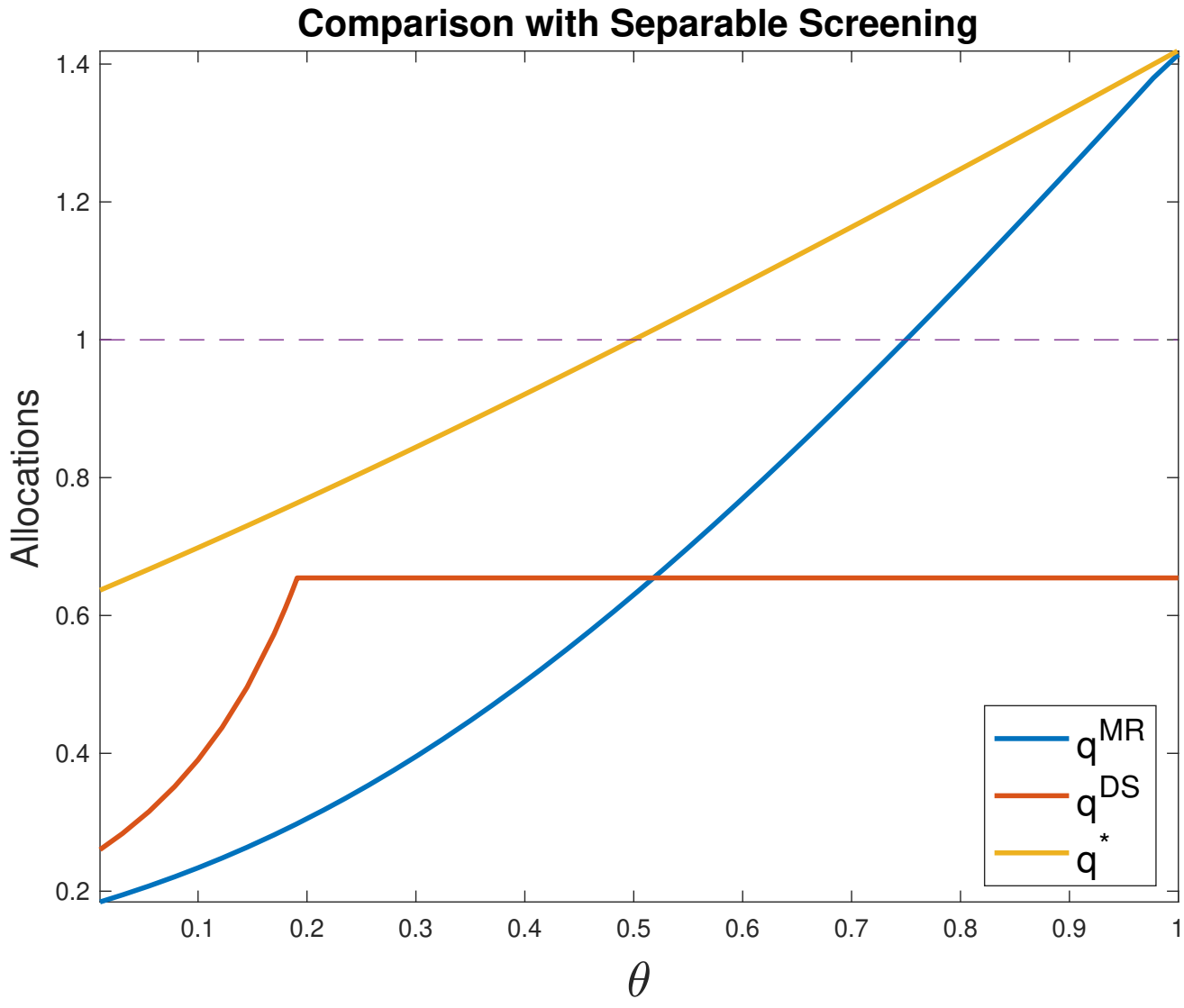
Necessary conditions for mixed-strategy equilibria:

Lemma 6

If $(H_1, \ldots, H_N)$ is a Nash equilibrium in mixed strategies with multiple active players, then for all i with $H_i(q) > 0$ for some $q > 0$ we have:

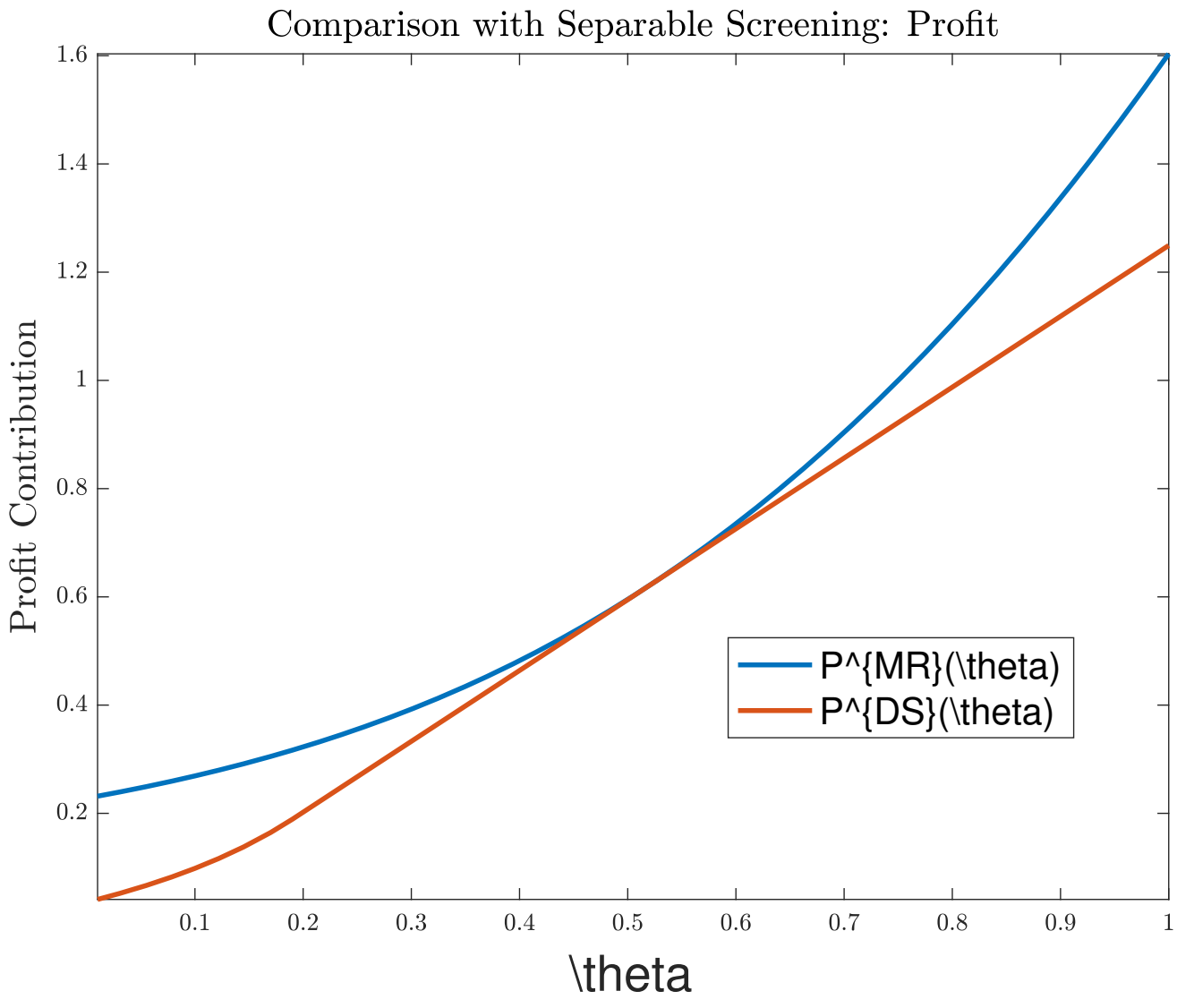
$$\prod_{j \in \mathcal{N} \setminus \{i\}} H_j(q) = \begin{cases} \frac{c'(q)}{V'(q)}, & \text{if } q \in (0, \overline{q}^M), \\ 1, & \text{if } q \geq \overline{q}^M. \end{cases}$$

Single buyer



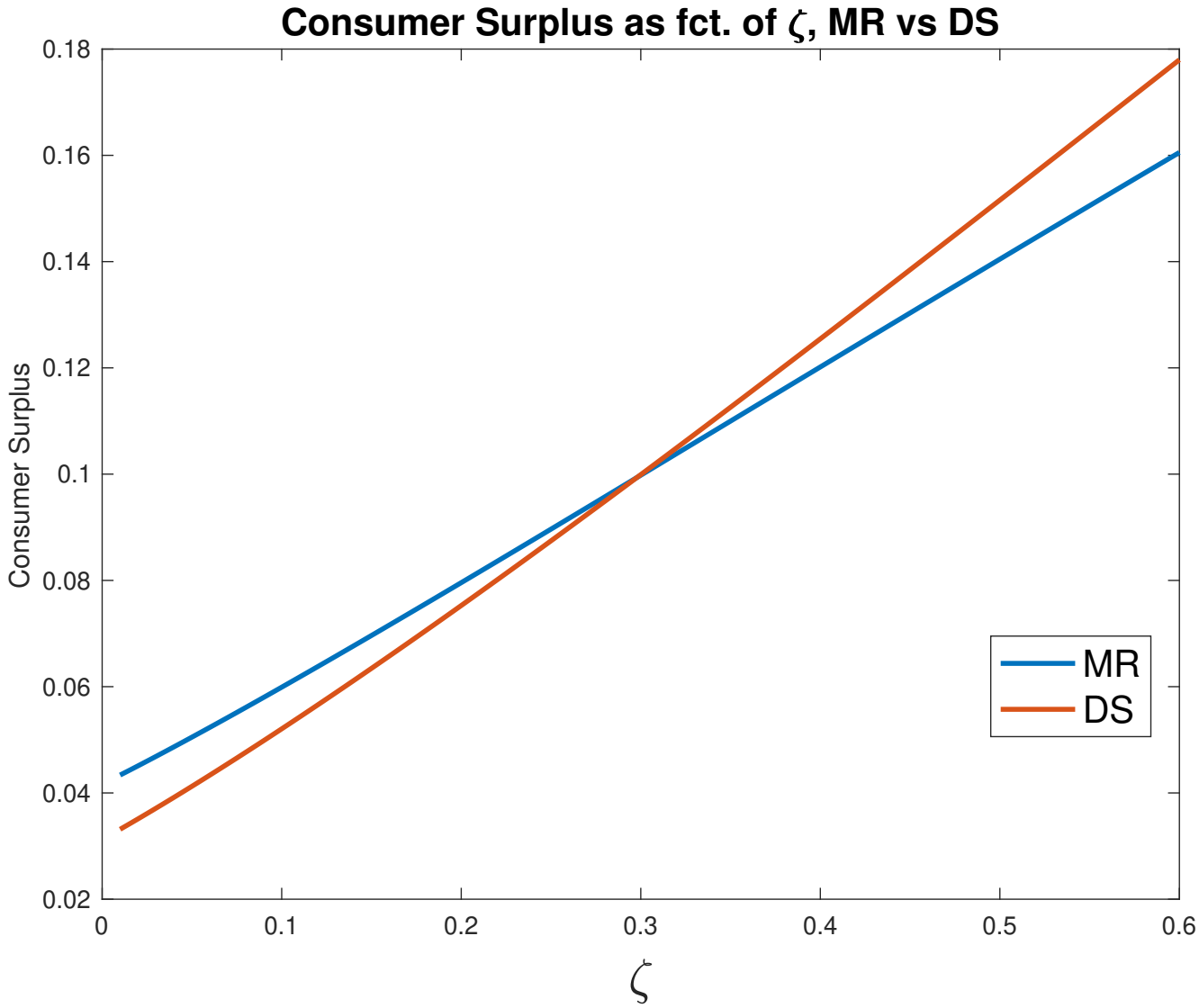
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Single buyer



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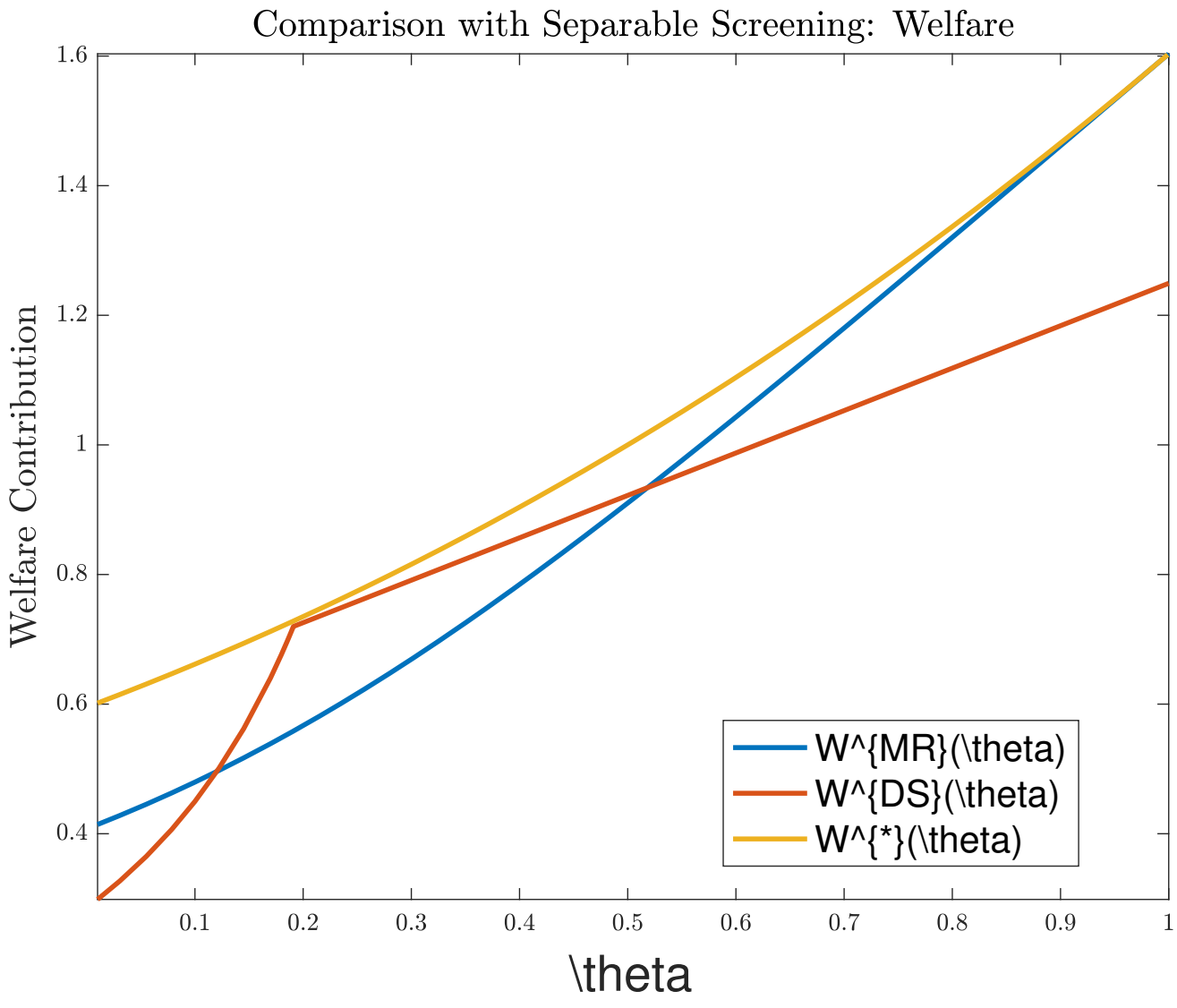
Single buyer



Consumer surplus of $\boldsymbol{q}$ is $\int_\Theta \boldsymbol{q}(\theta)(1 - F(\theta))\,\mathrm{d}\theta$, with

$$u(q, \theta) = \zeta g(q) + \theta q.$$

Single buyer



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